

Amortized Bayesian parameter recovery for dynamical systems with conditional flow matching

The `amortix` package: architecture, evaluation methodology, and benchmarks

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Abstract

We describe `amortix`, a package for amortized Bayesian estimation of the parameters of dynamical systems — ordinary and stochastic differential equations and discretized PDEs — from noisy observations at a finite set of time points. A conditional flow-matching model is trained once on simulated (parameter, data) pairs and thereafter produces posterior samples for any new observation set without likelihood evaluations or per-instance optimization; a single trained network handles any number and placement of observation points, the setting required for sequential experiment refinement and Bayesian experimental design. The package couples three components that we argue are only useful together: (i) a set-transformer encoder whose input representation is learnable rather than prescribed — a monotone mixture-of-scales reparametrization of the value axis, attention phases computed from physical observation times, and explicit set-cardinality features; (ii) a benchmark suite of fourteen systems spanning finance, neuroscience, epidemiology, ecology, pharmacokinetics, classical mechanics, and reaction–diffusion PDEs; and (iii) an evaluation harness in which simulation-based calibration is used only as a screen, and the instrument of record is comparison against exact or validated Markov chain Monte Carlo reference posteriors on the same observed points. Trained once per system in this design-amortized mode, the default network matches exact or validated reference posteriors to median Fréchet distances (in reference-normalized coordinates) of 0.01–0.07 on five of six reference-equipped systems — evaluated across design sizes with a fresh random design per test observation set, where it is comparable to or better than networks trained for each single design — while the single elevated entry, dense-design pharmacokinetics, is a measured effect: its posterior contracts to within a few multiples of the network’s absolute error floor, which stays at the $\text{few} \times 10^{-3}$ level of the prior range across systems whose posterior widths differ by $30\times$. Reaching those numbers required auditing the reference posteriors themselves — on the jump-diffusion, chains of the length we had been using disagreed with each other more than the model disagreed with them, and lengthening them moved the reported error from 0.19 to 0.05 — and residual errors decay steadily with the simulation budget. We compare against neural posterior estimation and flow-matching posterior estimation from the `sbi` package, both network families of BayesFlow 2, and a port of the Simformer, on three systems with exact or validated references. On a multiplicative process, all external configurations either inflate the volatility posterior by factors of 1.4–3.2 or become overconfident (widths 0.4–0.9) depending on the input coordinates they are given, while the learnable representation recovers the exact posterior from raw data. On an additive control process the same external tools improve markedly — the systematic bias disappears and the width inflation shrinks from 2.5–3.2 to 1.6–1.9 — which localizes the dominant failure to input conditioning while showing that a residual representation gap remains even in the benign case. We document this failure mechanism — the conditioning of the encoder input — in three distinct forms, together with learnable modules that repair each one.

1 Introduction

Recovering the parameters of a differential-equation model from noisy measurements of its solution is one of the basic inverse problems of applied science: a modeler writes down a dynamical system — an ODE, an SDE, or a PDE — whose right-hand side depends on unknown physical constants, observes a trajectory at a finite set of time points, and asks what the observations say about the constants. Like most inverse problems, this one is ill-posed in the practical sense: different parameter values can produce nearly indistinguishable observations, especially when the state is partially observed or the sampling is sparse,

so a point estimate both hides the ambiguity and is unstable under noise. The Bayesian formulation resolves this by reporting the full posterior distribution over parameters given the data — ambiguity becomes posterior spread and correlation, quantities one can act on — at the price of two computational obstacles. First, for most stochastic and partially observed systems, the likelihood of the observations requires integrating over unobserved trajectory segments and is intractable or expensive. Second, modern workflows need posteriors *repeatedly*: calibration studies process thousands of synthetic observation sets, sequential experiments update the posterior after every new measurement, and Bayesian experimental design compares candidate observation schedules by the posteriors they would produce — so per-instance MCMC or optimization inside such loops is the dominant cost even when a likelihood is available.

Simulation-based inference (SBI) addresses both issues by learning a conditional generative model of the parameters given the data from simulated pairs — parameters drawn from the prior, observation sets produced by the simulator — thereby amortizing the cost of inference into a one-time training phase (Cranmer et al., 2020). Neural posterior estimation (NPE) and its sequential variants (Papamakarios and Murray, 2016; Greenberg et al., 2019) fit normalizing flows by maximum likelihood; flow-matching and diffusion variants (Wildberger et al., 2023; Gloeckler et al., 2024) replace the density estimator with transport-based generative models. Mature software exists: the **sbi** toolkit (Boelts et al., 2024), BayesFlow (Radev et al., 2020), and the **sbibm** benchmark collection (Lueckmann et al., 2021).

This report describes **amortix**, a compact PyTorch package (about 3,500 lines; one default model of 4.8×10^5 weights) for amortized Bayesian parameter recovery in dynamical systems. Three aspects of its design distinguish it from the toolkits above, and the report is organized around measuring their consequences.

The first concerns the interface between raw data and the network. Conditional density estimators are conventionally fitted to a fixed-length data vector after global standardization; the estimator sees whatever coordinates the user chose. We show in Sec. 7.4 that on multiplicative and heavy-tailed processes this convention fails in a specific, reproducible way across four different generative engines from three codebases, and that supplying hand-derived coordinates does not repair it uniformly: with identical inputs, different implementations end up over-dispersed, overconfident, or biased. In **amortix** the transformation from observations to network inputs is itself part of the trained model — a monotone mixture-of-scales reparametrization of the value axis, observation times entering attention as continuous phases, and explicit design-size features (Sec. 3); each component is motivated by a measured failure and validated by an ablation.

The second concerns observation designs. **amortix** trains a single network to approximate $p(m | S)$ coherently for every subset S of the measurement points of an experiment — arbitrary sizes and placements, including spatio-temporal sensor designs for PDEs (Sec. 7.3). This is the object consumed by sequential refinement and Bayesian experimental design, and supporting it changes the training procedure (Sec. 4): designs must be resampled fresh at every optimizer step, and the distribution of design sizes becomes a modeling choice with measurable consequences.

The third concerns evaluation. Simulation-based calibration (Talts et al., 2018) is a necessary-condition test whose statistical power at practical sample sizes is limited; we measure that limit (Sec. 5) and use SBC only as a screen. The instrument of record throughout is the comparison of posterior mean and width against an exact or validated-MCMC reference posterior on the same observed points, and the package ships this machinery — closed forms, exact-likelihood adaptive Metropolis samplers, chain validation — as part of its public API. The concern motivating this discipline is documented in the SBI literature: learned posteriors can be confidently wrong in ways that rank diagnostics do not detect (Hermans et al., 2022).

The remainder of the report presents the method (Sec. 2.2), the architecture (Sec. 3), training procedures (Sec. 4), the evaluation methodology (Sec. 5), the benchmark suite (Sec. 6), experimental results including comparisons with **sbi**, BayesFlow, and the Simformer on three systems with exact or validated references (Sec. 7), the three input-conditioning problems the architecture solves (Sec. 8), and limitations (Sec. 9). Appendix A defines every benchmark system with its equations, priors, and observation model; Appendix G details the reference posteriors; Appendix H maps the terminology of this report to package options.

2 Method

2.1 Problem formulation

Let a dynamical system depend on a parameter vector $m \in \mathbb{R}^d$ with a known prior $p(m)$, uniform on a box in all benchmarks below. The system produces a state trajectory $x(t)$ — through an ODE $\dot{x} = f(x; m)$, an SDE $dx = f(x; m)dt + \sigma(x; m)dW_t$, or a spatially discretized PDE — and the trajectory is observed through a noisy channel at K points:

$$y_i = h(x(t_i)) + \varepsilon_i, \quad \mathcal{D} = \{(t_i, y_i)\}_{i=1}^K. \quad (1)$$

We call $\mathcal{D}$ the *observation set* of one experiment; it is the entire input to inference. Training and evaluation below involve many independently simulated observation sets, each paired with the parameter that generated it. The observation operator h may expose only part of the state: in the benchmarks below, only the price of an asset while its stochastic volatility is latent, only the membrane voltage of a neuron while its gating variables are latent, or only three spatial sensors of a PDE field. The target is the posterior $p(m | \mathcal{D}) \propto p(\mathcal{D} | m)p(m)$.

Three standing assumptions delimit the setting. The parameter is finite- and low-dimensional ($d = 2$ – 5 in every benchmark below) with a prior that factorizes across components and has tractable marginal CDFs (uniform boxes throughout; the construction extends to any such prior). The simulator can be run forward cheaply and repeatedly, but nothing else is required of it — no likelihood evaluations, no gradients. Observation times (and sensor indices) are known exactly; the measurement noise ε_i is independent across observations with a known per-system law.

Running example. The Ornstein–Uhlenbeck process is used throughout to make constructions concrete: $dX = -\theta X dt + \sigma dW$ with $m = (\theta, \sigma)$, prior box $[0.3, 3] \times [0.2, 1.5]$, stationary start $X_0 \sim \mathcal{N}(0, \sigma^2/2\theta)$. An observation set is, for instance, $K = 74$ noisy values of X on a fixed two-rate grid, or — in the variable-design setting — any K values at arbitrary times. Its posterior has the features that recur across the suite: the diffusion coefficient σ is identified sharply by high-frequency increments; the reversion rate θ only weakly over a finite horizon; and the width of both marginals must shrink as observations accumulate. An estimator can therefore be wrong in the posterior mean, in the posterior width, or in the width-versus- K trend, and the evaluation methodology of Sec. 5 is built to detect all three.

Although the posterior itself is intractable, sampling from the *joint* distribution of parameters and data is easy: draw $m_j \sim p(m)$, simulate the system, apply the observation channel. This yields arbitrarily many training pairs $(m_j, \mathcal{D}_j)$, and a conditional generative model trained on them approximates the posterior for every observation set simultaneously — the cost of inference is moved into a one-time training phase, and inference for a new observation set is a forward pass. We impose one requirement beyond standard amortization: a single trained network must remain correct for any number K and any placement of observation times (and, for PDEs, sensor positions). Formally, for every subset S of the measurement points of one experiment the network must approximate $p(m | S)$, and the family of answers must be coherent — in particular the posterior must widen as points are removed. We refer to this requirement as *design amortization*; it is the object consumed by sequential refinement and by Bayesian experimental design, where candidate observation schedules are ranked by the posteriors they would produce.

2.2 Posterior sampling by conditional flow matching

The idea of the method is to represent the posterior not by a density formula but by a *transport*: a learned, data-conditioned deformation that carries a fixed reference distribution onto the posterior. Sampling then amounts to drawing from the reference and applying the deformation — there is no normalizing constant to compute and no invertibility constraint to build into the network, because the deformation is defined implicitly as the flow of a velocity field, and the field itself is fitted by ordinary least-squares regression on simulated pairs. This construction is conditional flow matching (Lipman et al., 2023).

Concretely, parameters are first reparametrized componentwise,

$$m \mapsto \Phi^{-1}(F_{\text{prior}}(m)) \in \mathbb{R}^d, \quad (2)$$

where F_{prior} collects the marginal prior CDFs and Φ is the standard normal CDF. In the new coordinates the prior is exactly $\mathcal{N}(0, I)$ and box constraints disappear; usefully, the uninformative limit then costs nothing — when the data carry no information the posterior coincides with the reference and the transport

is the identity. From here on we work exclusively in these coordinates and keep writing m for the parameter in them; the inverse of (2) is applied exactly once, at the end of sampling, to return draws on the original scale.

The transport is indexed by a *virtual time* $\tau \in [0, 1]$ — an algorithmic variable with no relation to the physical time t at which the dynamical system is observed; t appears only inside observation sets, τ only in the transport. Subscripts in τ denote the position along the transport path: m_1 is a parameter drawn jointly with its observation set $\mathcal{D}$ (the endpoint), $m_0 \sim \mathcal{N}(0, I)$ is pure reference noise — not the transform of any parameter, but the raw material that the transport will have turned *into* a posterior draw by $\tau = 1$ — and between them lies the training interpolation $m_\tau = (1 - \tau)m_0 + \tau m_1$. The velocity network is trained to predict the direction of this path by regression:

$$\mathcal{L}(\theta) = \mathbb{E}_{m, \mathcal{D}, m_0 \sim \mathcal{N}(0, I), \tau \sim U[0, 1]} \|v_\theta(m_\tau, \tau, \mathcal{D}) - (m_1 - m_0)\|^2. \quad (3)$$

The minimizer of (3) is the marginal velocity field whose flow transports $\mathcal{N}(0, I)$ at $\tau = 0$ to $p(m_1 \mid \mathcal{D})$ at $\tau = 1$ (Lipman et al., 2023). Sampling draws $m_0 \sim \mathcal{N}(0, I)$, integrates $dm/d\tau = v_\theta(m, \tau, \mathcal{D})$ from $\tau = 0$ to 1, and maps the endpoint back through the inverse of (2); the numerical settings of the integration are implementation details, collected in Sec. 4. The same estimator family, with a different conditioning architecture, is available in `sbi` as FMPE (Wildberger et al., 2023); the comparison of Sec. 7.4 therefore isolates the effect of the input representation from the choice of generative engine.

No likelihood evaluations, simulator gradients, or per-instance optimization are required at any stage; the simulator is executed only forward, during training-set generation.

The training objective is consistent in the following sense.

Proposition 1 (consistency). *Let $(m, \mathcal{D}) \sim p(m)p(\mathcal{D} \mid m)$ and let m_1 be given by (2). If the loss (3) is minimized over all measurable velocity fields, then for $p(\mathcal{D})$ -almost every $\mathcal{D}$ the flow of the minimizer transports $\mathcal{N}(0, I)$ at $t = 0$ to the conditional law $p(m_1 \mid \mathcal{D})$ at $t = 1$; equivalently, the sampling procedure above draws exactly from the posterior (Lipman et al., 2023; Wildberger et al., 2023).*

The proposition is an idealization: with a finite network, a finite simulation budget, and a numerical solver, no calibration guarantee survives, and the residual error becomes an empirical quantity. Section 5 builds the instruments that measure it.

3 Conditioning architecture

The velocity field $v_\theta(m, \tau, \mathcal{D})$ must be conditioned on an observation set of time-stamped measurements, and three properties of that set drive the architecture: the number of observations K varies from experiment to experiment (by two orders of magnitude in the variable-design setting); the observation times are arbitrary real numbers, in general not equally spaced, so positional schemes built for uniform sequences do not apply; and the times carry all ordering information, so the output must not depend on how the observations happen to be stored. These properties rule out the fixed-length-vector conditioners that are the default in NPE implementations, and they are naturally met by a transformer operating on one feature vector per observation — throughout this section, a *token*.

3.1 Overall scheme

The network evaluates $v_\theta(m, \tau, \mathcal{D})$, where m is the normalized parameter vector partway along its transport path — the interpolation m_τ at training time, the current ODE state at sampling time — and τ is the virtual time of the transport. The design follows the encoder–decoder pattern that is standard in conditional flow-matching and diffusion models: the data are encoded once, and a decoder combines that encoding with the object being generated,

$$v_\theta(m, \tau, \mathcal{D}) = \text{Dec}(m, \tau, \text{Enc}(\mathcal{D})), \quad (4)$$

with explicit types: `Enc` maps the observation set — the K time-stamped measurements of one experiment — to a memory of K vectors, $\text{Enc}(\mathcal{D}) \in \mathbb{R}^{K \times d_{\text{model}}}$, computed once per observation set, and `Dec` : $\mathbb{R}^d \times [0, 1] \times \mathbb{R}^{K \times d_{\text{model}}} \rightarrow \mathbb{R}^d$ combines the current transport point and the virtual time with that memory. Two differences from the image and text instances of this pattern dictate the concrete form (Fig. 1). First, the conditioning is not a fixed-size embedding but an observation set whose size K varies: `Enc` is therefore a transformer over per-observation tokens, its output is a *memory* of K encoded states rather than one pooled vector, and `Dec` reads the memory by cross-attention — the one standard

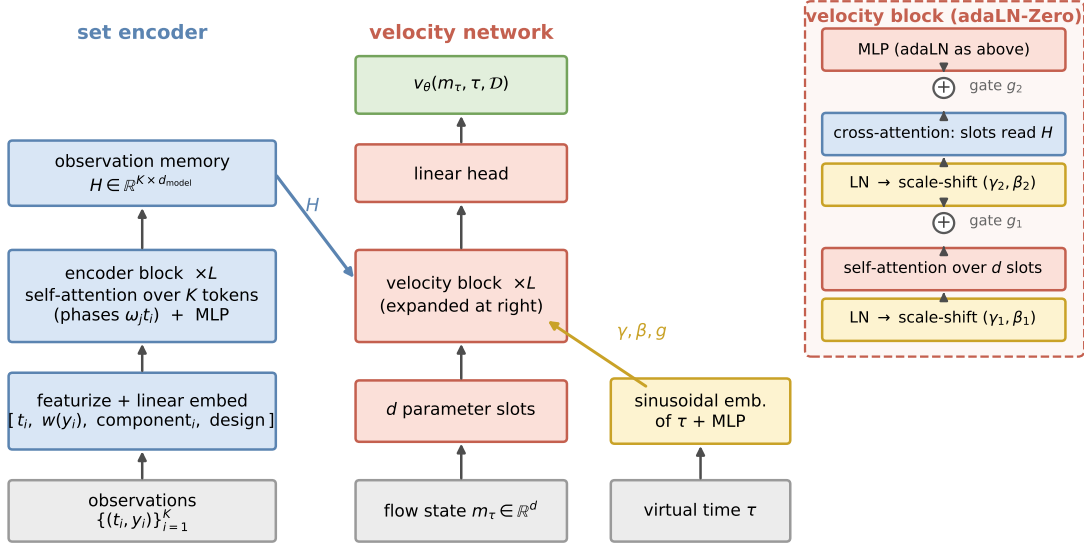


Figure 1: Model architecture; arrows denote data flow, bottom to top. Left: each observation is featurized and linearly embedded into one token; L transformer encoder blocks, whose attention phases are computed from the physical observation times (Sec. 3.2), produce a memory H of K encoded states. Middle: the d components of the flow state m_τ form d parameter-slot tokens processed by L velocity blocks and a linear head into the velocity v_θ . Right: one velocity block expanded — self-attention across parameter slots, cross-attention reading the observation memory H , and an MLP, each preceded by layer normalization with scale-shift (γ, β) and followed by a gated residual connection (g , marked $+$); the triples (γ, β, g) are produced from the virtual time τ by a zero-initialized map, so the untrained block is the identity (Peebles and Xie, 2023).

mechanism indifferent to K . Second, the generated object is not an image but a short vector of d parameters, so the decoder is small: the d components of m are treated as d tokens, self-attention among them expresses posterior correlations (the ridge of the Fisher–KPP posterior, the drift–volatility coupling of GBM), and the virtual time τ enters every block through zero-initialized adaptive layer normalization (adaLN-Zero), the conditioning mechanism of diffusion transformers (Peebles and Xie, 2023), so that the untrained network is an identity-like map.

On the encoder side, observation i is described by a few features — its time t_i , its transformed value $w(y_i)$ with w the learnable monotone map of Sec. 3.3, an index identifying the sensor or state component that produced the value (multivariate states contribute one token per observed component per time), and the design-size feature of Sec. 3.3 — and lifted to the model width by a learned linear map. The attention mechanism that lets these tokens interact is the subject of the next subsection; the choice of features is the subject of Sec. 3.3.

This scheme was not the first attempt but the survivor of a sequence of measured alternatives; the failed ones are retained in the package as reproducible controls and are compared under a common protocol in the ablation study of Appendix E. All sizes are configuration parameters. One default configuration (width 64, four heads, three blocks per side) is used unchanged for every experiment in this paper; at the training-set sizes we use, models of this scale proved sufficient — doubling width and depth changes no benchmark result (Sec. 7.5) — and nothing in the construction caps the scale.

3.2 Attention phases from observation times

We adapt rotary position embeddings — the standard positional scheme of modern transformers (Su et al., 2024) — to irregular time sampling.

First the classical construction. For token states $x_1, \dots, x_K \in \mathbb{R}^{d_{\text{model}}}$ entering an attention layer, each head computes

$$q_i = W_Q x_i, \quad k_i = W_K x_i, \quad u_i = W_V x_i, \quad W_Q, W_K, W_V \in \mathbb{R}^{d_h \times d_{\text{model}}}, \quad (5)$$

and token i aggregates the others with weights $a_{il} = \text{softmax}_l(\hat{q}_i^\top \hat{k}_l / \sqrt{d_h})$, where $\hat{q}, \hat{k}$ are position-modulated: each query and key is split into $d_h/2$ planar blocks $q_i^{(j)} \in \mathbb{R}^2$ and block j is rotated by

an angle set by a per-token scalar ϕ_i ,

$$\hat{q}_i^{(j)} = R(\omega_j \phi_i) q_i^{(j)}, \quad \hat{k}_i^{(j)} = R(\omega_j \phi_i) k_i^{(j)}, \quad R(\phi) = \begin{pmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{pmatrix}. \quad (6)$$

In the classical scheme the scalar is the token’s integer position in the sequence, $\phi_i = n_i$, and the frequencies form a geometric ladder across blocks, $\omega_j = \omega_{\max}(\omega_{\min}/\omega_{\max})^{(j-1)/(d_h/2-1)}$, so that high-frequency blocks resolve neighboring positions and low-frequency blocks long ranges. Because rotations preserve inner products up to the relative angle,

$$\hat{q}_i^\top \hat{k}_l = \sum_{j=1}^{d_h/2} (q_i^{(j)})^\top R(\omega_j(\phi_l - \phi_i)) k_l^{(j)}, \quad (7)$$

every attention logit depends on the positions only through differences $\phi_l - \phi_i$ — relative position awareness with no positional parameters.

For irregularly sampled data the integer position is meaningless, and the generalization is immediate: we set $\phi_i = t_i$, the physical time stamp of observation i , and choose the frequency range so that the periods $2\pi/\omega_j$ span from the finest observable time gap to several times the horizon. Nothing else changes; by (7), every logit now depends on the time stamps only through differences $t_l - t_i$. The substitution of physical time for position is the same one that time-aware attention models for irregular clinical and event series make in other positional schemes (Horn et al., 2020; Shukla and Marlin, 2021). Two consequences follow at once: the encoder is exactly permutation-invariant (relabeling tokens leaves the right-hand side of (7) unchanged; verified numerically to 10^{-8}), and variable designs require no positional bookkeeping. The phases enter only the encoder’s self-attention; the cross-attention from parameter slots to the observation memory is unmodified attention, since parameter slots carry no time stamp and the memory states already encode temporal structure.

Integer positions, by contrast, create an implicit train/test contract on token placement: a model trained on spread-out subsampled positions and evaluated on contiguously packed ones acquires posterior biases of +1.1 to +3.3 standard deviations while every component remains individually correct (Appendix E). Physical-time phases eliminate the contract rather than manage it.

3.3 Input features

The remaining design choices concern what the tokens contain. Three of them matter and are described in turn: a learnable transform of the observed values, an optional increment-based token construction for directly observed diffusions, and an explicit design-size feature. Each occupies a few lines of code; each is load-bearing, as the ablations of Appendix E show.

Value transform. State variables of the benchmark systems live on very different scales, and some — prices, concentrations, populations — are multiplicative: the information sits in relative rather than absolute changes, and the coordinate in which such a variable is well-behaved is logarithmic. Rather than prescribing coordinates per problem, each value enters its token through a learned monotone map,

$$w(x) = \sum_{k=1}^S a_k \operatorname{sign}(x) \log(1 + |x|/s_k), \quad a_k \geq 0. \quad (8)$$

Each basis function is linear for $|x| \ll s_k$ and logarithmic for $|x| \gg s_k$, so the family contains the identity, the signed logarithm, and every compression in between; positivity of the weights makes w monotone by construction; and — the operative property — every candidate compression is present from initialization, so training only reweights fixed nonlinearities, and the downstream network can exploit whichever coordinate helps from the first optimizer step. The scales s_k sit on a fixed geometric grid covering the dynamic range of the data ($S = 8$ octaves by default, one setting for all fourteen systems, no per-problem adjustment).

This structure has to earn its place against simpler-looking learnable alternatives, and we tested them on GBM, where the correct coordinate (log-price) is known and the cost of missing it is a measured +0.48 sd volatility bias. A generic two-layer MLP on the raw values, and a random-Fourier-feature map, both left the bias in place at matched budget: a free-form map must discover the shape and the scale of the compression simultaneously, from gradients arriving through the whole downstream network, and in practice it does not. A classical one-parameter statistical family — the Yeo–Johnson power transforms

(Yeo and Johnson, 2000), in essence $x \mapsto ((1 + |x|)^\lambda - 1)/\lambda$ applied with sign symmetry, which passes through the identity at $\lambda = 1$ and tends to the logarithm as $\lambda \rightarrow 0$ — also failed, in an instructive way: trained end-to-end, λ never left its initialization, because a single global exponent must traverse a gradient plateau before the downstream network has learned anything that would reward the move. The mixture (8) removes the bias reliably, and on GBM the trained weights concentrate on the small-scale components: the model learns the log-price coordinate.

Increment features for directly observed diffusions. For scalar SDEs whose state is observed directly — no latent components — the package offers a second token construction: after sorting by time, consecutive observations contribute

$$[t_{(i)}, w(y_{(i)}), \Delta w_i, (\Delta w_i)^2, w_t(\Delta t_{(i)})], \quad \Delta w_i = w(y_{(i+1)}) - w(y_{(i)}).$$

The motivation is elementary: for a Markov observed process the likelihood factorizes over consecutive pairs, transition densities of diffusions depend on increments and gaps, and squared increments carry the quadratic-variation information that identifies diffusion coefficients. This is an inductive bias of the same character as convolutions for images — useful where its assumption holds, not a requirement of the method: the point-token path of Sec. 3.1 is fully general, and the ablations of Appendix E quantify what the increment tokens buy at fixed budget.

The rule for choosing between the constructions — increment tokens exactly when the observed process is Markov — was tested in both directions: an increment-token model class transfers across Markov systems without adjustment (GBM to OU), and on the Heston model, whose observed price is not Markov because volatility is latent, increment and point tokens perform identically across all 25 evaluation cases, as the factorization argument predicts. The package selects the construction from a per-problem flag.

Two further normalization details matter for processes with jumps, where squared increments within a single observation set span several orders of magnitude and a global feature normalization becomes unstable — concretely, on the Merton benchmark the running normalization scale is set by rare jump outliers and the resulting representation drift produces a +1.12 sd volatility bias. First, increment features are logarithmically compressed before normalization; this is a near-identity on systems without jumps (since $\log(1 + x) \approx x$ for small x) and by itself turns the Merton failure into a calibrated posterior across all 25 evaluation cases (Sec. 8). Second, the normalization is made adaptive to the observation set: robust summary statistics s of the observation set, pooled over non-padding positions, are mapped by a zero-initialized MLP to per-channel scale and shift corrections,

$$h_c \mapsto (1 + \gamma_c(s)) h_c + \beta_c(s) \tag{9}$$

— a feature-wise affine modulation, FiLM in the terminology of Perez et al. (2018). Its measured contribution is specific and modest: it removes a residual +0.27 sd bias that survives on the densest Merton designs (the effective jump-detection threshold shifts with the unknown volatility), and it does so as well as a hand-crafted per-observation-set rescaling, within error bars ($+0.13 \pm 0.06$ vs $+0.09 \pm 0.05$ sd; Sec. 8). Three conditions are required for it to help: zero initialization, modulation applied *after* the running normalization, and pooled statistics that exclude padding positions.

Design size. Posterior width is governed by the number of observations — in regular models the error scales as $K^{-1/2}$ — but K is a global property of the set, and softmax attention, which computes normalized weighted means, is nearly invariant to it. Softmax attention computes normalized weighted means, which are nearly invariant to the number of tokens — yet posterior width is governed by K . The deficit is measurable by linear probes from the pooled encoder output: all architecture variants of Table 10 recover posterior-mean statistics ($R^2 = 0.83\text{--}0.94$), while $R^2(\log K)$ ranges from 0.985 to 0.498 and ranks the variants exactly in the order of their width errors. The repair is to make the count available, and the package default is the simplest form: $\log K$ enters every token as one of the input features. An alternative that requires no explicit feature — appending a few learned auxiliary tokens to the set, in the spirit of register tokens (Darcet et al., 2024), and letting the model accumulate the count in them — achieves the same effect in our ablations; the number of such tokens is a hyperparameter with no measurable influence beyond one. With the count available, the width error on the hardest case ($K = 100$) drops from 1.95 to 1.49 with bias unchanged; the residual is closed by the training procedure of the next section.

4 Training procedures

Common settings. Sampling integrates the velocity field with a fixed-step midpoint scheme, 20 steps by default; the step count was fixed once by a convergence check (fourth-order Runge–Kutta with three times as many steps changes no benchmark statistic), and evaluation uses an exponential moving average of the training weights. The default optimizer is Muon (Jordan et al., 2024) — the momentum of every two-dimensional weight is orthogonalized by a Newton–Schulz iteration before it is applied, with Adam retained for biases, normalizations, and embeddings — and its step size follows the batch-size rule $\eta(B) = \min(10^{-2}\sqrt{B/512}, 2 \times 10^{-2})$; the measurements behind the rule are collected in Appendix B. Adam at base learning rate 3×10^{-4} remains available (`optimizer="adam"`) and is the recipe of the fixed-design tables in Appendix C (batch size 64). Under either optimizer: cosine schedule with linear warmup, gradient clipping, and evaluation on an exponential moving average of the weights with decay 0.999. At every optimizer step the virtual times τ are drawn afresh (stratified over $[0, 1]$) together with fresh reference noise m_0 , so no pairing between observation sets and virtual times is ever repeated. For fixed-design problems the training set consists of n simulated observation sets and training runs for a fixed number of epochs over it; the gallery of Sec. 7.2 uses $n = 40,000$ simulations and 35 epochs.

Design amortization. Training data are simulated once on a fine grid; observation designs are then resampled *from the stored fine trajectories at every optimizer step*. The procedure has three components, each fixed by a controlled comparison on three systems (GBM, Ornstein–Uhlenbeck, FitzHugh–Nagumo):

1. *Fresh designs at every step.* Re-drawing each trajectory’s observation subset at each step makes design memorization impossible and lets one simulation serve arbitrarily many designs. The alternative — one fixed design per trajectory — closes the dense-design error at long budgets but overfits the design set, producing overconfidence at sparse K (width ratio 0.86 against the exact reference).
2. *The design-size law.* K is drawn 50% log-uniform on $[K_{\min}, K_{\max}]$ and 50% uniform on the dense half of the range. Log-uniform sampling alone puts too little mass at large K and leaves dense designs under-trained; the mixture removes the effect.
3. *Budget scaling.* When higher accuracy is required, simulations and optimizer steps are scaled together at constant epochs; Sec. 7.5 shows the residuals then follow power laws in this single budget variable, and Sec. 7.5 measures why the scaling must be joint: passes over reused observation vectors eventually destroy accuracy, while fresh simulations do not.

With this procedure, the GBM posterior width matches the exact per-design posterior to 5–7% uniformly over $K = 5, 9, 20, 100$.

Budgets and stopping. Step budgets are set with deliberate slack, and the shipped weights are chosen by validation: training monitors a validation set — built by the same machinery as the test evaluation sets, from disjoint seeds — and replaces the final weights with the best monitored checkpoint only when the improvement exceeds the monitor’s own resolution and the monitored value itself sits at least three floors above it — a monitor asked to rank values of its own resolution returns noise in both directions. With that margin in place, overshooting a budget costs time, never quality, so the step budget is not a per-system quantity to tune. The measurements behind the margin — including the two failure modes of the naive best-checkpoint rule — are in Appendix B.

Package defaults for Algorithms 1–2: the step rule $\eta(B)$ of Sec. 4, monitor period $\Delta = T/24$, resolution multiple $\kappa = 3$ (Appendix B), and $L = 20$ ODE steps, fixed once by the convergence check of Sec. 4.

5 Evaluation methodology

Two instruments with different roles are used throughout, plus a diagnostic probe.

5.1 Simulation-based calibration as a screen

SBC (Talts et al., 2018) tests a necessary condition: for $m^* \sim p(m)$ and $\mathcal{D} \sim p(\mathcal{D} \mid m^*)$, the rank of each component of m^* among posterior samples must be uniform. We run SBC per parameter (uniformity test on 500 simulated observation sets with 200 posterior draws each unless stated; $p > 0.05$ passes) and report coverage of central 50% and 90% intervals.

Algorithm 1 Training a design-amortized posterior

Require: simulator, prior $p(m)$; simulation budget n , batch size B , step budget T

Require: validation set V with monitor floor φ ; monitor period Δ ; resolution multiple κ

Require: step rule $\eta(B)$ and schedule (Sec. 4)

```
1: draw  $m_i \sim p(m)$  and simulate full fine-grid trajectories  $X_i$ ,  $i = 1 \dots n$ 
2: initialize the posterior network;  $(\theta^*, F^*) \leftarrow (\emptyset, \infty)$ 
3: for  $t = 1 \dots T$  do
4:   sample a batch of trajectories; for each, draw a fresh design: size  $K$  from the mix law, then  $K$ 
     random (time, sensor) points; tokenize
5:   draw virtual times  $\tau$  (stratified) and reference noise  $m_0 \sim \mathcal{N}(0, I)$ ; set  $z_\tau \leftarrow (1 - \tau) m_0 + \tau \tilde{m}$ 
6:   Muon step at  $\eta(B)$  on  $\|v_\theta(z_\tau, \tau, \text{Enc}(\text{tokens})) - (\tilde{m} - m_0)\|^2$ ; update the EMA weights  $\bar{\theta}$ 
7:   if  $t \bmod \Delta = 0$  then
8:      $F \leftarrow$  score of  $\bar{\theta}$  on  $V$ ; if  $F < F^*$  then  $(\theta^*, F^*) \leftarrow (\bar{\theta}, F)$ 
9:   end if
10: end for
11: return  $\theta^*$  if  $F_{\text{last}} - F^* > \varphi$  and  $F^* \geq \kappa \varphi$ ; else the final  $\bar{\theta}$ 
```

Algorithm 2 Inference: n posterior draws for one observation set S

Require: trained network (Enc, v_θ); observation set S (any size, any placement); sample count n ; ODE step count L ; $h \leftarrow 1/L$

```
1: tokenize  $S$ ; encode once:  $C \leftarrow \text{Enc}(\text{tokens})$ 
2: draw  $z^{(1)} \dots z^{(n)} \sim \mathcal{N}(0, I)$  ▷ or the learned base, where enabled
3: for  $\tau = 0, h, \dots, 1 - h$  do ▷ midpoint rule; all  $j$  batched
4:    $z^{(j)} \leftarrow z^{(j)} + h v_\theta(z^{(j)} + \frac{h}{2} v_\theta(z^{(j)}, \tau, C), \tau + \frac{h}{2}, C)$ 
5: end for
6: return denormalized  $z^{(1)} \dots z^{(n)}$ 
```

The power of the test at this size is limited, and because our conclusions depend on knowing that limit, we measured it: on cases where an exact reference is available, SBC at 300–500 observation sets fails to reject posterior width ratios up to ≈ 1.3 , and its per-parameter p -values are unstable between retrainings when residual biases are in the range 0.1–0.2 posterior standard deviations. SBC also cannot penalize a posterior that ignores the data and returns the prior; the prior-mean and ridge controls of Sec. A exist for that reason. SBC is therefore used as a cheap screen, never as a verdict.

5.2 Reference posteriors as the record

Wherever a tractable likelihood exists, we compare distributions on the same observed points. The headline statistic is a single number per observation set, the squared Fréchet distance between Gaussian fits of the two sample sets in reference-normalized coordinates (each parameter divided by the reference posterior standard deviation),

$$d_F^2 = \|\mu_a - \mu_r\|^2 + \text{tr}(\Sigma_a + \Sigma_r - 2(\Sigma_r^{1/2} \Sigma_a \Sigma_r^{1/2})^{1/2}) \quad (10)$$

— the construction used as the FID score in the image-generation literature, and we adopt that name. Two normalizations of the same construction are used and always named: FID_σ divides each coordinate by the *reference posterior standard deviation* (the default — an unadorned FID in any table means FID_σ), and FID_Δ divides by the *prior range*. They answer different questions — error relative to what the data determine versus error relative to the scale of the problem — and a model of fixed absolute accuracy reads very differently in the two wherever the posterior is narrow (Sec. 7.3).

FID is zero for a perfect match, dimensionless, and combines the three ways a posterior can be wrong: location, width, and orientation error. For scale, a bias of 0.1 reference-sd in each of d dimensions gives FID $0.01d$, and a posterior 30% too wide in one dimension gives FID ≈ 0.09 . Where the direction of a failure matters — too wide is merely wasteful, overconfident is silently dangerous — we additionally quote the per-parameter decomposition: the signed posterior-mean error in reference-sd units (*bias*) and the ratio of standard deviations (*width*; 1 exact, above one conservative, below one overconfident).

References are of three kinds, with construction and validation detailed in Appendix G: closed form for the linear-Gaussian benchmark and geometric Brownian motion (conjugate on the log-returns); an

system	type	d	obs.	design	reference posterior
linear-Gaussian	linear map	4	full	fixed	exact Gaussian
Ornstein–Uhlenbeck	SDE	2	full	fixed	exact-likelihood MCMC
geometric Brownian motion	SDE	2	full	fixed	conjugate closed form
Cox–Ingersoll–Ross	SDE	3	full	variable	exact-likelihood MCMC
double well	SDE	3	full	fixed	—
stochastic Lotka–Volterra	2-d SDE	4	full	fixed	—
FitzHugh–Nagumo	2-d SDE	4	v only	fixed	—
SEIR	4-d SDE	5	I, D	fixed	—
polynomial-drift SDE	SDE	5	full	fixed	—
Heston	2-d SDE	5	price only	variable	—
Merton jump-diffusion	jump SDE	5	full	variable	Poisson-mixture MCMC
Hodgkin–Huxley	4-d ODE	4	V only	variable	—
pharmacokinetics (Bateman)	ODE (closed form)	3	concentration	variable	log-normal MCMC
Fisher–KPP	PDE (64-point grid)	2	3 sensors	variable	PDE-likelihood MCMC

Table 1: The benchmark suite. “obs.” lists the observed coordinate when the state is only partially visible; “design” distinguishes the fixed-grid gallery from variable-design systems, where every observation set carries its own set of (time, sensor) observation points. Dashes mark systems with no validated reference: these are evaluated by the calibration screen only (Appendix A), which is precisely the regime that motivates carrying reference-equipped systems in the same suite.

exact-likelihood adaptive Metropolis sampler for Ornstein–Uhlenbeck, Merton, pharmacokinetics, and Fisher–KPP, whose likelihoods are exact per-gap transition densities (one likelihood evaluation is one PDE solve in the Fisher–KPP case); and, for every MCMC reference, validation by a second chain from a well-separated start. A reference is used only once its two chains agree, and the between-chain posterior-mean discrepancy — in posterior-sd units — is the floor below which nothing can be measured, quoted alongside the model’s value throughout. That floor scales with chain length: the Merton jump-diffusion needs 2×10^5 -draw chains to clear it (two chains then agreeing to 0.016 sd in the median, 0.060 at worst), and systems whose chains do not converge at any affordable length are reported under the calibration screen instead.

Resolution of the estimator. Both sample sets are finite, so (10) is positive even for a perfect posterior, and the size of that offset sets what the metric can resolve. We measured it directly: drawing *two independent sample sets from the same exact GBM posterior* and scoring one against the other gives a median FID of 0.0157 at 500 samples, 0.0038 at the 2,000 used throughout this report, 0.0009 at 8,000, and 0.0002 at 32,000 — a clean $1/n$, as the estimation error of the fitted means and covariances predicts. Two consequences. Differences below ≈ 0.005 at $n = 2,000$ are not resolved, so values in that range are reported as “at the resolution limit” rather than compared. And the best values in this report sit above it rather than at it: the design-amortized GBM model reads 0.0107 at $n = 2,000$, 0.0100 at 8,000, and 0.0072 at 32,000 on the same evaluation set, so after subtracting the estimator offset its excess over the reference is ≈ 0.007 at every sample size — a real error of the model, not an artifact of the ruler.

Equation (10) is the squared 2-Wasserstein distance between Gaussian fits; the division by the reference standard deviation is ours, since the parameters carry different physical units. Distribution-free discrepancies and the classifier two-sample test of Lueckmann et al. (2021) rank the systems identically (Appendix F); the choice with consequences is the normalization, which the notation above makes explicit.

6 Benchmark suite

The suite contains fourteen systems (Table 1); Appendix A gives each one with its equations, parameter ranges, and observation model. Variable-design training — observation subsets resampled at every optimizer step — is the package default and the recommended mode: beyond enabling inference at any design, the resampling acts as data augmentation over designs. Nine systems are nevertheless benchmarked on conventional fixed grids, for two practical reasons: their closed-form or cheap references live on those grids, and the external tools of Sec. 7.4 consume fixed-length inputs.

All SDEs are integrated by Euler–Maruyama on the stated fine grid; ODEs by RK4 (Hodgkin–Huxley by a semi-implicit scheme for the gating variables); the PDE by finite differences. Systems marked

random designs are trained and evaluated in the design-amortized mode of Sec. 2: the observation times (and, for the PDE, sensor indices) of every observation set are drawn at random, and one trained network serves all of them; the remaining systems use one conventional fixed grid. The *classical estimator* listed for each system is the standard non-Bayesian method a practitioner would apply; all of them return point estimates, not distributions, and they serve as an accuracy floor for the posterior mean. Three recur and are defined here once. *Kramers–Moyal regression* estimates the drift and squared diffusion from the conditional first and second moments of the observed increments ($f(x) \approx \mathbb{E}[\Delta X \mid X = x]/\Delta t$, $\sigma^2(x) \approx \mathbb{E}[\Delta X^2 \mid X = x]/\Delta t$) and fits the parametric form to these estimates by least squares. *Multi-start nonlinear least squares* fits the deterministic solve of the system to the observed trajectory by minimizing squared error over parameters, restarted from many random initial points to escape local minima. *Exact maximum likelihood* is available where the transition density is known in closed form. Fixed-design observations are token sets over the union of the stated channels; “fast/slow” channels observe every step over a short window and every j -th step over the horizon respectively. Priors are uniform on the stated boxes.

7 Experimental results

This section is organized as an experimental study: the protocol — references, baselines, model sizes, budgets — is fixed first; the main results (posterior overlays and the global metric on every system) follow; then the comparison with existing tools, the scaling behavior in budget, model size and data reuse, and the evaluation of design amortization.

7.1 Experimental protocol

References and baselines. Every accuracy number in this section is the FID of Sec. 5.2 against a reference posterior computed on the same observed points, reported as the median over the system’s evaluation set: a frozen collection of observation-set instances — 200 where the reference is closed form, 100 where it is cheap MCMC, 32 where each reference costs minutes of validated MCMC — built once with two-chain validation (Sec. 5.2) and reused for every number in this report, so any two entries differ only in the model being scored. The reference samplers and their cost per observation set for 2,000 samples (laptop CPU; validation, where used, doubles it): the exact conjugate posterior for GBM (<1 ms), exact-likelihood MCMC for GBM and OU (42–124 ms), mixture-likelihood MCMC for Merton and pharmacokinetics (0.03–0.1 s), and PDE-likelihood MCMC for Fisher–KPP (271 s, every likelihood evaluation being a PDE solve). The external amortized tools appear in the same tables as our models, at the shared 20,000-simulation budget of Sec. 7.4. Simulation-based calibration serves as the screen for systems without a validated reference; its per-parameter results for the fixed-design gallery are in Appendix A.

Models and training. For each reference system we train the same architecture at three sizes — tiny (0.1M parameters), the default small (0.5M), and big (2.3M) — plus two miniature sizes (nano, 26k; pico, 7k) used to locate the capacity floor. All runs follow the procedure of Sec. 4. (The Adam-recipe tables of Appendix C carry one coupling of their own: at width 128 the Adam learning rate is halved, otherwise optimizer instability masquerades as a capacity effect; the default recipe has no such coupling — Appendix B.) Simulation budgets range from 5,000 to 480,000 per system. The size ladder under the default recipe opens the results (Table 2); the size $\times$ budget grid shared with the external tools is Table 6 (Appendix C). Observation designs are random at training and at evaluation wherever the system defines a design distribution; fixed designs appear only where the interface of an external tool or reference requires them (Sec. 7.4, linear-Gaussian check).

7.2 Main comparison: quality, model size, data, and baselines

One comparison, run identically on every task, opens the results: the package’s default recipe — Muon with the batch step rule, batch 512, a deliberately generous 90,000 steps with validated stopping (Sec. 4) — trained at five model sizes on every system with a validated random-design reference, three seeds at the top three sizes, and scored on the frozen evaluation sets. The rows are the systems admitting a validated random-design reference at evaluation-set scale — including Cox–Ingersoll–Ross, which joined once its closed-form transition density (noncentral χ^2) was wired into the same machinery, with an exact simulator and gap likelihood describing the same chain. The linear-Gaussian check keeps its per-design

system	floor	FID _σ at model size (parameters)				
		pico (7k)	nano (26k)	tiny (98k)	small (0.5M)	big (2.3M)
geometric Brownian motion	0.0018	0.0064	0.0039	0.0036 ± 0.0004	0.0028 ± 0.0001	0.0035 ± 0.0006
Ornstein–Uhlenbeck	0.0013	0.0297	0.0281	0.0294 ± 0.0026	0.0282 ± 0.0031	0.0281 ± 0.0017
Cox–Ingersoll–Ross	0.0028	0.0786 ± 0.0178	0.0823 ± 0.0131	0.0733 ± 0.0110	0.0809 ± 0.0071	0.0700 ± 0.0036
pharmacokinetics	0.0023	0.0846	0.0319	0.0340 ± 0.0021	0.0262 ± 0.0062	0.0251 ± 0.0024
Merton jump-diffusion	0.0065	0.0663	0.0299	0.0219 ± 0.0007	0.0247 ± 0.0016	0.0267 ± 0.0050
Fisher–KPP	0.0100	0.0705 ± 0.0161	0.0415 ± 0.0033	0.0377 ± 0.0054	0.0307 ± 0.0036	0.0282 ± 0.0035

Table 2: The main result: FID_σ against validated references (median over the evaluation set), one design-amortized network per cell, every cell trained with the package default recipe at a budget of 120,000 simulations and evaluated on the frozen evaluation sets of Sec. 7.1 ($K = 20$; Merton $K = 56$; Fisher–KPP $K = 40$, whose reference chains are 4,000 draws each — every likelihood evaluation is a PDE solve — hence its higher floor). Cells with $\pm$ give the mean and half-range over independent training seeds. Every underlying record carries the recipe and package fingerprint that produced it.

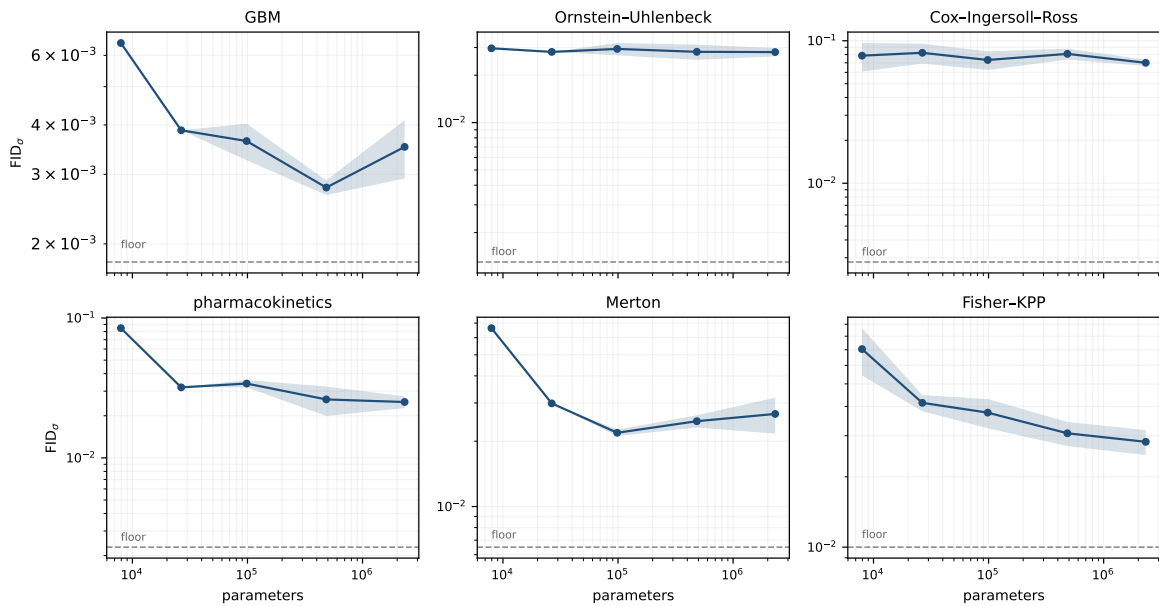


Figure 2: The capacity ladder of Table 2 drawn against each evaluation set’s resolution floor (dashed): median FID_σ versus parameter count, band spanning the seed range where three seeds were run. Both axes logarithmic.

reference in the design-size record (Table 3); the remaining seven systems of Table 1, where no reference has yet passed two-chain validation, are covered by the calibration screen. Table 2 is the result; Fig. 4 shows the corresponding posteriors.

The table reads in four statements. On GBM every size from tiny up sits within two floors of the evaluation set’s own resolution (0.0031–0.0036 against 0.0018), so those cells are at the measurement limit and are not compared further. On pharmacokinetics the ladder is monotone from 0.085 at pico to 0.025 at big — the system that was the suite’s pathology under earlier recipes is now its cleanest capacity curve. Merton saturates already at tiny (0.022; the larger sizes differ by less than the seed spread). And Ornstein–Uhlenbeck is size-independent at ~ 0.028 — twenty floors above the evaluation set’s resolution — so its residual is a property of the task, not of capacity or recipe. Fisher–KPP is monotone across the whole ladder (0.071 $\rightarrow$ 0.028), and Cox–Ingersoll–Ross is flat at every size at 0.07–0.08 — twenty-five floors, OU-like: a residual owned by the task. Validated stopping is load-bearing here, not a formality: the best-on-validation checkpoint replaced the final weights in over half the runs — in 10 of 11 on Cox–Ingersoll–Ross, whose validation curve turns as early as a sixth of the budget — under a double condition stated entirely in the monitor’s units: the improvement must clear the monitor’s floor, and the monitored value itself must sit at least three floors up, because a monitor asked to rank values of its own resolution returns noise in both directions (Appendix B). Fig. 2 draws the six ladders against their floors; Fig. 3 overlays the one-dimensional posterior marginals of the median-error test instance of every system against

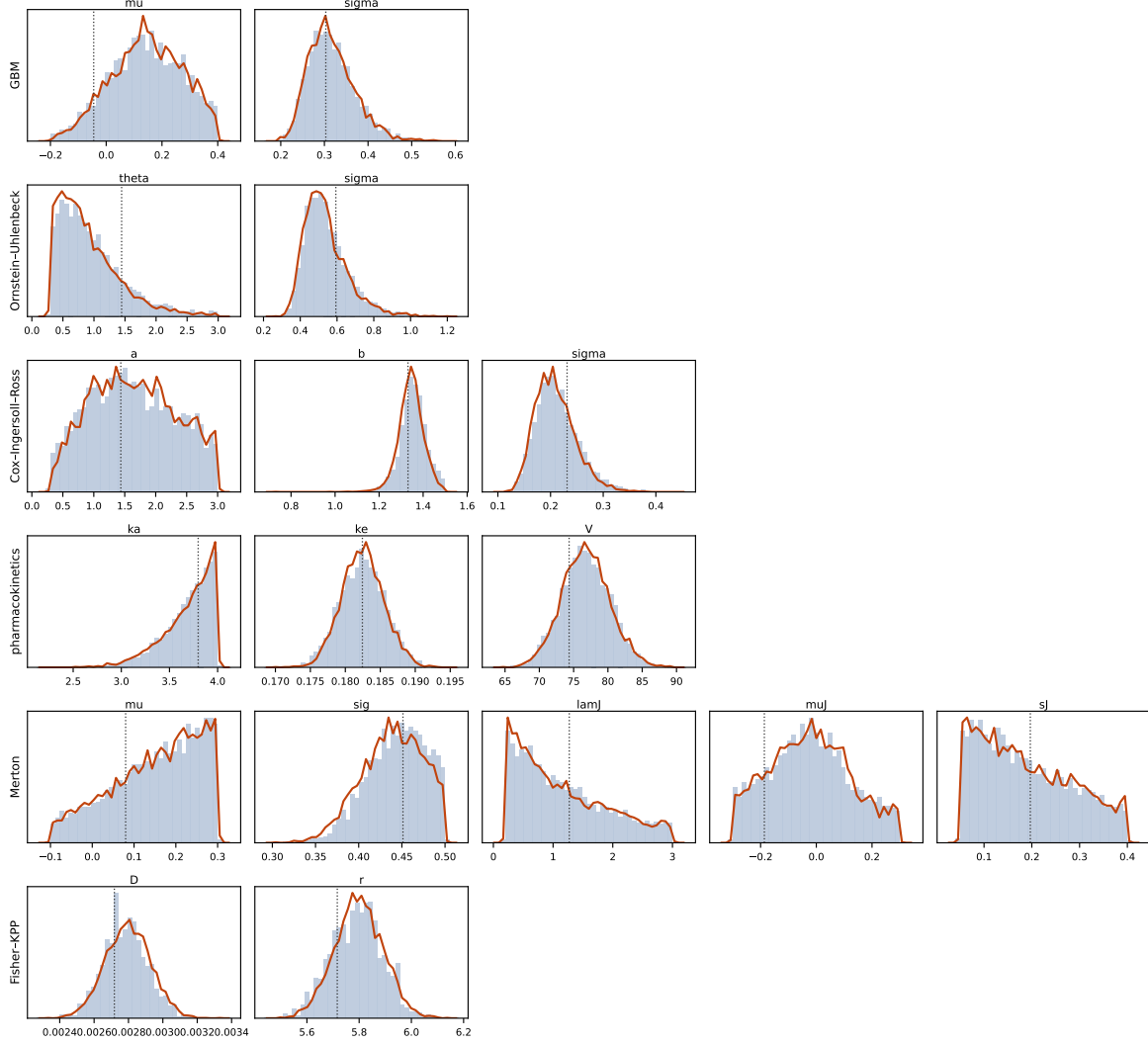


Figure 3: One-dimensional posterior marginals, model (line) against reference (filled), for the median-FID_σ test instance of each system in Table 2; the big model, 4,000 draws each; vertical line: the true parameter. The median instance is chosen so the figure shows typical, not best-case, agreement.

its reference. The dependence on the simulation budget is Fig. 5; the size $\times$ budget grid shared with the external tools is Table 6 (Appendix C).

At the shared 20,000-simulation budget, every column of the grid beats every external tool on every task it enters — by one to two orders of magnitude on GBM and OU on raw data, and by 2–6 $\times$ on Fisher-KPP — and the gap survives hand-engineering the external tools’ inputs; Sec. 7.4 dissects where it comes from.

Figure 4 shows the object the package produces — the joint posterior itself. For every system with an exact or validated reference, the posterior of the trained model is overlaid on the reference for one representative test observation set — three systems with simple posteriors that serve as checks (linear-Gaussian, GBM, OU), and three whose posterior geometry is nontrivial: the Merton jump-diffusion, where volatility and jump rate compete to explain the same variance; pharmacokinetics at six irregular blood draws, the data-poor clinical regime; and Fisher-KPP, whose posterior concentrates on the curved ridge $Dr \approx \text{const}$ fixed by the observable front speed. The learned posteriors reproduce location, scale, orientation, and — where present — the curvature of the ridge.

Table 3 reports the accuracy record across design sizes; the systems with no validated reference are covered by the calibration screen (Appendix A). Every reference row is evaluated in the package’s default mode — one network per system, a fresh random design per test observation set — and reads median FID 0.01–0.07 on five of the six systems, the five-parameter Merton jump-diffusion included once its reference is computed with converged chains. The single elevated entry is pharmacokinetics (0.13), whose posterior

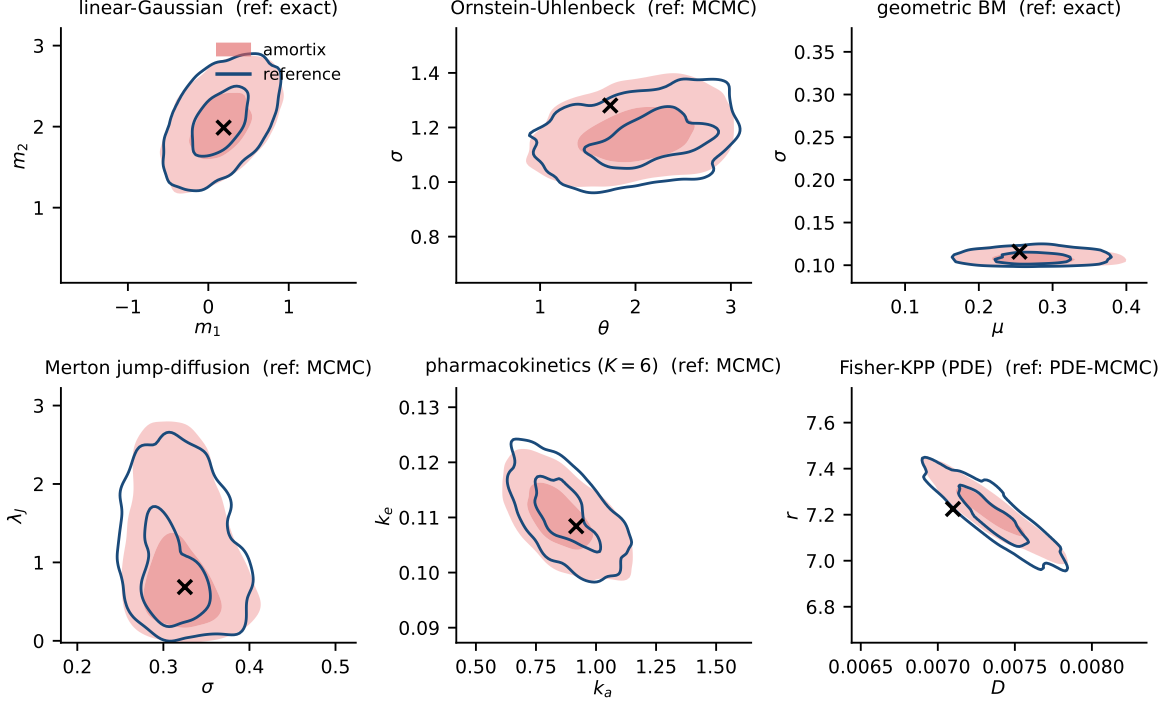


Figure 4: Joint posteriors of `amortix` (filled contours: 39% and 86% highest-density regions) against exact or validated-MCMC reference posteriors (open contours, same levels) on one test observation set per system; crosses mark the true parameters. Top row: systems with closed-form or cheap exact references. Bottom row: systems where the reference itself required exact-likelihood MCMC with two-chain validation — including the ridge-shaped Fisher-KPP posterior, where the model reproduces the curvature, not just the moments.

at dense designs contracts to within a few multiples of the network’s absolute error floor (Sec. 7.3). Per-parameter screen results for the systems without references are in Appendix A.

Pharmacokinetics is the exception, and its mechanism is dissected in Sec. 7.3: its posterior is the narrowest in the suite, so the network’s absolute error — comparable to every other system’s — reads as a large FID once divided by that width. Together with Table 3, this figure is the central evidence that one small default model recovers full posteriors across the suite.

7.3 Amortization over observation designs

The package’s central claim is that one trained network answers $p(m \mid S)$ for *every* observation design S — any number of points, any placement. Table 3 and Fig. 6 test the claim on every system of the suite where a per-design reference can be computed and validated — six of the fourteen: each test observation set draws a fresh random design, the reference posterior is recomputed on exactly those points, and FID is evaluated per design-size bucket. Wherever a fixed-design specialist of the same architecture and simulation budget exists, the single amortized network is comparable to it at the specialist’s own design size (GBM 0.018 vs 0.012; OU 0.023 vs 0.029; linear-Gaussian 0.046 vs 0.039; Fisher-KPP 0.050 vs 0.036) and is the only one of the two that answers anywhere else on the K axis. Amortization over designs is not a trade-off against accuracy at any point we measured; it also removes a failure mode, since resampling the observation subset at every optimizer step doubles as data augmentation (Sec. 7.5 measures what the fixed-design procedure pays for reusing identical tokens across passes). The two systems with elevated FID are not noise; the two paragraphs after the table take them apart.

Why FID_σ rises with the design size. Pharmacokinetics is the one system whose FID_σ *rises* as the design gets denser ($0.09 \rightarrow 0.30$ from $K = 5$ to 50). The mechanism separates cleanly in the two normalizations of Sec. 5.2, and both factors were measured on the same evaluation sets rather than inferred. Write $w_\Delta(K)$ for the reference posterior width and $\varepsilon_\Delta(K)$ for the network’s absolute posterior-mean error, both in prior-range units. Densifying the design contracts the posterior: w_Δ falls from $(0.27, 0.027, 0.063)$ at $K = 5$ to $(0.053, 0.0070, 0.023)$ at $K = 50$ for (k_a, k_e, V) — factors of 2.8–5.1,

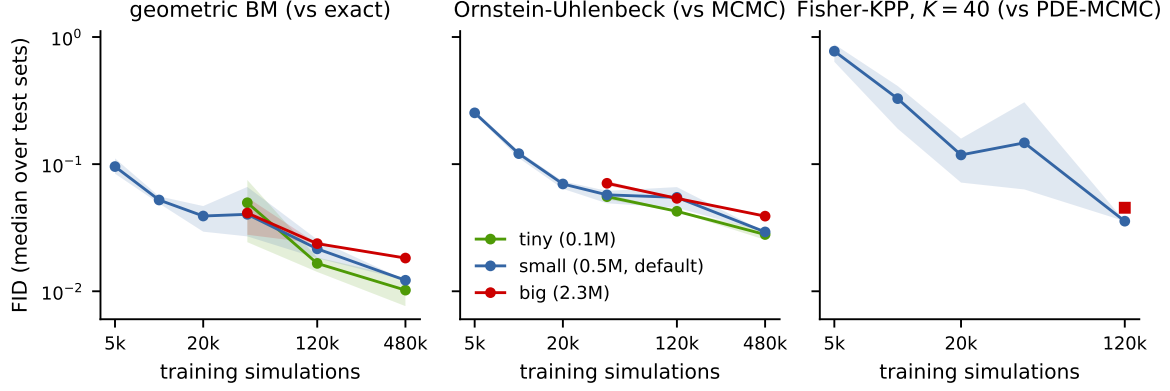


Figure 5: Median FID versus simulation budget at three model sizes (both axes logarithmic; optimizer steps scale with the budget at constant epochs; the big model uses the width-scaled learning rate). Lines: seed means; bands: seed ranges where multiple seeds were run. The widening Fisher-KPP band at intermediate budgets is seed-level freedom along the weakly identified ridge $\ln(r/D)$, which contributes little to the loss; refining the sampling ODE solver from 20 midpoint steps to 60 RK4 steps changes nothing measurable. On the two-parameter systems (left, middle) the three sizes coincide within seed spread — the budget axis, not the parameter axis, buys accuracy. Fisher-KPP (right): default size, with the big model at 120,000 marked by the square. References: exact conjugate posterior (GBM), exact-likelihood MCMC (OU), PDE-likelihood MCMC (Fisher-KPP, fixed $K = 40$ design).

system	K : sparse/medium/dense	FID $_{\sigma}$ at those K			specialist at its own K
		sparse	medium	dense	
geometric Brownian motion	5 / 20 / 100	0.006	0.008	0.018	0.012 ($K=95$)
Ornstein-Uhlenbeck	5 / 20 / 100	0.072	0.045	0.023	0.029 ($K=74$)
linear-Gaussian	2 / 4 / 6	0.012	0.035	0.046	0.039 ($K=6$)
Fisher-KPP	10 / 40 / 90	0.073	0.050	0.126	0.036 ($K=40$)
Merton jump-diffusion	5 / 20 / 100	0.027	0.046	0.057	—
pharmacokinetics	5 / 20 / 50	0.069	0.125	0.339	—

Table 3: The accuracy record: one design-amortized network per system, evaluated at design sizes never fixed in advance — each test observation set draws a fresh random design and its reference posterior is recomputed on exactly those points; entries are medians per design-size bucket. Last column: a separate network of the same architecture and simulation budget trained on a single fixed design, which exists only at its own K . GBM and OU are evaluated on the denser K -grid of Fig. 6; the Merton and pharmacokinetics rows use the converged references of Sec. 5.2. The pharmacokinetic rise with K is the precision floor dissected in Sec. 7.3.

bracketing the $\sqrt{10} \approx 3.2$ of a regular model. It does not move the network’s error: ε_{Δ} stays at $1\text{--}12 \times 10^{-3}$ across the whole range, with no trend in K . The bias term of FID $_{\sigma}$ is $\sum_i (\varepsilon_{\Delta,i}/w_{\Delta,i})^2$: a flat numerator over a shrinking denominator, and it duly grows from 0.025 to 0.121. The model is not becoming miscalibrated — its posterior widths match the reference to 1–8% at every design size — and the rise is not an artifact of the normalization either: a classifier two-sample test, which carries no scale convention, separates model from reference at 0.64 accuracy on the dense-design evaluation set against a null of 0.50. The network has an absolute localization floor of order $10^{-3}\text{--}10^{-2}$ of the prior range; dense designs resolve the true posterior past that floor, and FID $_{\sigma}$ reports the shortfall in units of what the data determine.

This is why the pharmacokinetic entries are reported per design size rather than pooled. Across the 32 test observation sets the values are bimodal — sets whose reference posterior is narrowest read several times worse than the rest — so a median over one 32-instance evaluation set moves between 0.04 and 0.15 between otherwise identical training runs, and no single number summarizes the system honestly. The stochastic systems never enter this regime: diffusion noise keeps their posteriors an order of magnitude wider than the error floor even at $K = 100$, and Fisher-KPP approaches it only at its densest bucket (0.126 at $K = 90$). The general caveat stands for any scale-relative metric: it reports a model of fixed absolute accuracy as increasingly poor wherever the posterior it is compared against is narrow.

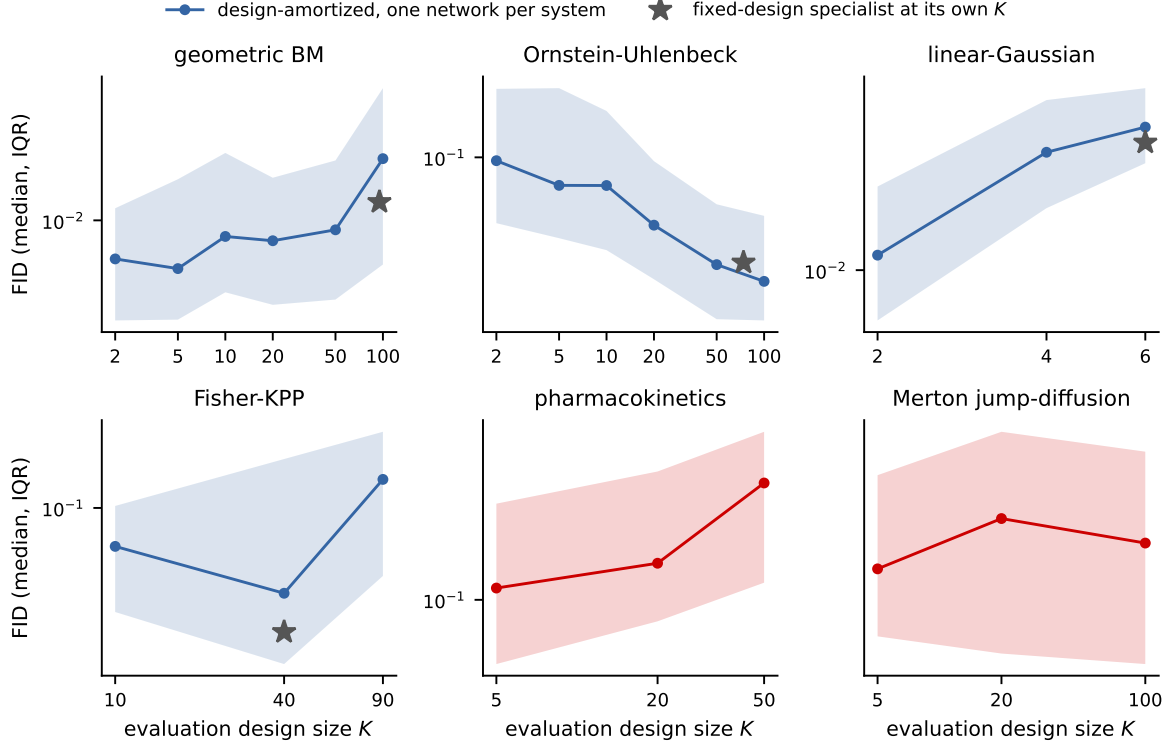


Figure 6: FID as a function of the evaluation design size, for one design-amortized network per system: median (line) and interquartile range (band) over the test observation sets of each bucket (100 per bucket for GBM, OU, and linear-Gaussian, 32 for the MCMC-referenced systems), each set with a fresh random design and a per-design reference. Stars: fixed-design networks of the same architecture and simulation budget, which exist only at their own K . Both axes logarithmic; note the different vertical scales.

Where references end. One further system — Hodgkin–Huxley — is a deterministic flow with Gaussian observation noise, so its likelihood is tractable in principle and a per-design reference looks within reach. It is not, and the reason is worth recording. Under deterministic dynamics the posterior concentrates on needle-thin sets, and adaptive random-walk chains do not converge on such geometry: independent chains land 1.8–18 sd apart, against the 0.22 sd worst case of every validated reference in this report. By the validation standard of Sec. 5.2 the reference is rejected, and the system remains under the calibration screen. The precision-floor argument above predicts that a valid reference, if computed by a stronger sampler, would find elevated FID_σ here as well.

Variable-design zoo. With the class rule and training procedure fixed, six further systems were trained without per-system adjustments: Heston (five parameters, latent volatility; increment tokens give no advantage over point tokens, consistent with the class boundary of Sec. 3.3); Merton, whose dense-design failures are cured by the heavy-tail treatment of Sec. 8; Hodgkin–Huxley; pharmacokinetics, which at $K = 6$ irregular blood draws shows bias -0.01 to -0.06 sd and widths 1.09–1.33 against exact-likelihood MCMC; and Fisher–KPP, whose posterior concentrates on the ridge $D \cdot r \approx \text{const}$ with the along-ridge bias decaying with budget (Sec. 7.5). Under the default recipe the per-parameter screen reads 4/4 on Hodgkin–Huxley and 3/5 on Heston, where the leverage correlation fails at $p = 4 \times 10^{-4}$.

Which encoder components make design amortization work is an architecture question, separate from the accuracy claims above; the controlled comparison of eight encoder variants that fixed the design of Sec. 3 is in Appendix E.

7.4 Comparison with existing tools

Protocol. We compare against the amortized estimators of maintained packages: NPE with its default masked autoregressive flow (Papamakarios et al., 2017) and FMPE (Wildberger et al., 2023) from `sbi` 0.27 (Boelts et al., 2024); the coupling-flow and flow-matching networks of BayesFlow 2.0 (Radev et al., 2020); and a port of the Simformer (Gloeckler et al., 2024), the closest architecture to ours, built from its

system	method	train	inference	FID
GBM (exact conjugate)	amortix CFM	236 s	70 ms	0.04
	sbi NPE / FMPE	170–215 s	36–359 ms	4.8 / 3.8
	BayesFlow 2 coupling / FM	790–1252 s	0.3–95 ms	2.6 / 2.3
	Simformer port	703 s	1049 ms	8.5
	<i>best external, hand-derived inputs</i>	—	—	0.34
Ornstein–Uhlenbeck (exact-lik. MCMC)	exact-likelihood MCMC	—	42 ms (CPU)	reference
	amortix CFM	226 s	77 ms	0.07
	sbi NPE / FMPE	114–134 s	19–279 ms	2.7 / 1.2
Fisher–KPP ($K = 40$, PDE-lik.)	exact-likelihood MCMC	—	124 ms (CPU)	reference
	amortix CFM	591 s	69 ms	0.07
	sbi NPE / FMPE	163–235 s	19–703 ms	0.5 / 76
	PDE-likelihood MCMC	—	271 s (CPU)	reference

Table 4: All methods at package defaults, one H200 GPU, equal 20,000-simulation budget; FID is the median over 200 (GBM), 100 (OU) and 32 (Fisher–KPP) test observation sets. Inference time is per observation set for 2,000 samples. The hand-derived row is the best external arm when given log-returns, the exact coordinates of the GBM likelihood factorization (full ladder: Appendix D). On Fisher–KPP every reference likelihood evaluation is a PDE solve (271 s of MCMC per observation set), so the entire training of the amortized model costs about as much as references for three observation sets.

authors’ minimal example. All arms get the same 20,000 simulations from the same prior and simulator, run at package defaults, and are compared on training cost, accuracy, and inference cost. Because these estimators consume fixed-length vectors, the comparison uses fixed designs — the only experiments in this report that do, and the reason the variable-design capability of Sec. 7.3 is not exercised here. Three systems separate the possible explanations of a gap: a multiplicative process where input conditioning is hostile (GBM, exact conjugate reference), an additive one where it is benign (Ornstein–Uhlenbeck), and a PDE where every likelihood evaluation is a forward solve (Fisher–KPP), which inverts the cost relationship between reference and amortized inference.

What the gap is made of. On raw data every external configuration misses the posterior by one to two orders of magnitude more than **amortix**, and the dominant mechanism is the input representation rather than the generative engine: **sbi** itself warns about precision loss while z-scoring GBM prices. Hand-derived coordinates help but do not converge to a common answer — with identical log-return inputs the external tools land between 0.34 and 1.5 and fail in different directions, NPE over-dispersed (σ width 1.44), both BayesFlow arms overconfident (widths 0.42–0.85, the silently dangerous direction). The additive control isolates the cause: on OU, where no floating scale exists, the external bias becomes consistent with zero and width inflation drops from 2.5–2.6 to 1.6–1.9, so conditioning accounts for the dominant part of the failure and a smaller representation-learning gap for the rest. Fisher–KPP shows the same defects can be discontinuous across parameters: FMPE at defaults fails outright on the diffusivity (+7.3 sd bias at $12.5\times$ width) while remaining reasonable on the reaction rate. Every defect here is invisible without a reference and lies within or near the blind region of SBC at practical sizes (Sec. 5). Specialists could improve each arm with problem-specific embedding networks; the comparison measures exactly what is delivered when the input representation is not hand-engineered per problem, and the Simformer arm in particular should be read as representative of its minimal recipe rather than of the method’s ceiling.

7.5 Data reuse and the capacity ladder

The grid of Table 6 scales optimizer steps with the simulation count at constant epochs. Two further grids separate what a budget buys when that convention is broken: passes over stored simulations versus fresh ones, and parameters versus data at five model sizes.

Fresh simulations beat repeated passes. Table 7 varies the number of simulated trajectories and the number of epochs independently, for the fixed-design GBM and OU models at the default size. Along constant compute — 10,000 trajectories $\times$ 120 epochs versus 120,000 $\times$ 10, equal optimizer steps — fresh data wins by an order of magnitude or more in FID. Passes are not merely diminishing but

destructive: at 120 epochs every configuration is worse than at 35, including the largest training set (OU: $0.066 \rightarrow 0.648$). The mechanism is memorization of the tokens: virtual times and reference noise are redrawn at every step (Sec. 4), so what repeats across passes is only the fixed observation vectors. Consistent with this, redrawing a random 75% token subset at every optimizer step (`token_dropout`) largely repairs the 120-epoch column — OU $1.89 \rightarrow 0.25$ at 10,000 trajectories, $0.65 \rightarrow 0.14$ at 120,000 — while slightly hurting the one cell that had barely overfitted (GBM at 120,000: $0.13 \rightarrow 0.28$): training on subsets while evaluating on the full design is its own train/test mismatch, so subsampling is a repair, not a free improvement. The design-amortized procedure performs the same augmentation natively and without the mismatch — every optimizer step sees a fresh design drawn from the law used at evaluation — which is why the amortized network of Table 3 is at least as accurate as its fixed-design counterpart at the same budget.

The capacity floor moves with the parameter dimension. Table 8 trains five sizes (7k–2.3M parameters) on four systems at a fixed budget of 120,000 simulations. The pattern is a floor that rises with the parameter dimension: at $d = 2$ the ladder is flat within seed spread; at $d = 3$ accuracy degrades several-fold below $\sim 10^5$ parameters; at $d = 5$ it improves monotonically up to the default size and then saturates — the two largest Merton entries differ by less than the reference’s own floor. Above the capacity floor the data axis outbids the parameter axis: quadrupling the simulations moves the default Merton model from 0.052 to 0.035, more than the twenty-four-fold parameter step at fixed data buys. Pharmacokinetics is not scaled past 120,000 simulations: its evaluation-set median at 480,000 swings between 0.04 and 0.24 across step counts with no monotone trend, too noisy a measurement to place on a grid.

Cost. A saturated model trains in 19–63 minutes per system on one H200 GPU (120,000 simulations); at the 20,000-simulation budget of the comparisons above, training takes 6–12 minutes. The same run on a laptop CPU takes roughly four times longer and produces a bit-identical posterior, because all stochastic generation is kept on a CPU random number generator. Drawing 2,000 posterior samples costs 426 ms per observation set on the laptop CPU and 72 ms batched (135 ms single) on the GPU. The small external models of Sec. 7.4 train and sample faster at their defaults; the tiny-model regime is kernel-launch-bound on GPUs (`sbi`’s default flow trains $6\times$ slower there than on the laptop CPU).

8 Conditioning of the encoder input

Encoder probing. To separate encoder failures from flow failures, we fit ridge regressions from the pooled encoder output to known sufficient statistics of the problem. High probe R^2 with poor posterior accuracy localizes the defect downstream of extraction. In the Merton analysis of Sec. 8, the failing configuration extracted robust volatility statistics *better* than the healthy one ($R^2 = 0.967$ on truncated realized variance), redirecting the repair from the encoder’s capacity to the stability of its input representation.

The learnable input representation of Sec. 3.3 exists to solve one problem in three forms: the conditioning of the numbers presented to the set encoder. The same problem accounts for the external failures of Table 9. We give each form with its measurement and its repair; all three repairs are learnable modules rather than problem-specific transformations.

Form 1: floating multiplicative scale. On GBM, raw price tokens force the network to divide squared increments by squared levels internally; with global standardization this manifests as a volatility bias of +0.48 posterior sd (confirmed against the closed-form posterior) that no amount of training removes. The repair is the monotone mixture reparametrization of Sec. 3.3, which learns the log coordinate; the Yeo–Johnson ablation there shows that what decides the outcome is the parametrization, not the hypothesis class. External estimators exhibit the same form on the same data (Table 9, raw-price rows).

Form 2: cardinality blindness. Normalized attention erases set size, which controls posterior width under design amortization; the probe measurements and the log K /register-token repairs are given in Sec. 3.3, with the arm-by-arm consequences in Table 10.

Form 3: heavy tails. On the Merton jump-diffusion, squared increments have sample standard deviation 3.1×10^4 against a diffusion signal of order 10^{-2} –1; running feature normalization never converges,

and the flow trains against a drifting representation, producing a $+1.12$ sd volatility bias. Encoder probing shows the information is extracted — the failing configuration encodes robust volatility statistics better than the healthy one — so the repair targets representation stability: logarithmic compression of increment features (near-identity on clean diffusions since $\log(1+x) \approx x$ for small x) restores all 25 evaluation cases. A precision follow-up against the Poisson-mixture reference then isolated a smaller ($+0.27$ sd) residual whose mechanism is a jump-classification threshold that floats with the unknown volatility; a per-observation-set set-conditioning module (pooled statistics modulating feature channels, initialized to the identity) removes it as effectively as a hand-crafted per-observation-set rescaling ($+0.127 \pm 0.055$ vs $+0.094 \pm 0.054$ sd, statistically indistinguishable). The module is zero-initialized and acts after feature normalization; both are required, and pooled statistics exclude padding positions.

9 Discussion and limitations

The evidence assembled here supports a specific claim: for dynamical-systems inverse problems, the input representation of an amortized posterior estimator is not a preprocessing detail but the dominant accuracy factor, and it can be made part of the model rather than part of the problem specification. The comparison of Sec. 7.4 shows the cost of the alternative — fixed representations with global standardization — across four generative engines, on a problem where the correct answer is known in closed form. The OU control separates the mechanisms: on additive data the bias of the external estimators disappears and their width inflation drops by roughly a factor of two but does not vanish, so the conditioning pathology accounts for the dominant part of the failure and a smaller representation-learning gap accounts for the rest.

Limitations. (i) The remaining inaccuracies of the suite decay as measured power laws in training compute (Fig. 5); we have not searched for methods that improve the exponents. (ii) The power limits of SBC imply that small residuals can only be certified where an exact or validated reference exists; systems without tractable likelihoods inherit the weaker screen, and carrying reference-equipped systems alongside them in one suite is our current mitigation. (iii) The comparison of Sec. 7.4 uses package defaults; it measures the out-of-the-box regime, not the ceiling of any method. (iv) Bayesian experimental design — the intended consumer of design amortization — is not yet demonstrated; the training procedure of Sec. 4 was built so that the required object $p(m | S)$ is already coherent across subsets S . (v) No real-data study with posterior predictive checks is included yet. (vi) The polynomial-drift system uses a dense coefficient prior; the scientifically interesting variant is a *sparse* prior — train on coefficient vectors with random supports and test whether the posterior recovers the support — which requires extending the normalization (2) to priors with atoms at zero, a planned extension of the package. (vii) All priors are uniform on boxes; the probit normalization (2) extends to any factorized prior with tractable CDFs, but correlated priors would require a different normalization.

A Benchmark systems and calibration detail

Linear-Gaussian. $y = Am + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, 0.5^2 I_6)$, with a fixed correlated design matrix $A \in \mathbb{R}^{6 \times 4}$; $m \in [-3, 3]^4$. Exact Gaussian posterior; the classical estimator is the posterior mean itself. This system checks the flow engine in isolation from the encoder.

Ornstein–Uhlenbeck. $dX = -\theta X dt + \sigma dW$, stationary start $X_0 \sim \mathcal{N}(0, \sigma^2/2\theta)$; $dt = 0.02$, 500 steps; channels: fast (1×50), slow (20×24). Prior $\theta \in [0.3, 3]$, $\sigma \in [0.2, 1.5]$. Classical: exact MLE from per-gap Gaussian transitions.

Geometric Brownian motion. $dS = \mu S dt + \sigma S dW$, $S_0 = 1$; $dt = 0.01$, 500 steps (five “years”); channels: fast (1×60), slow (10×40). Prior $\mu \in [-0.2, 0.4]$, $\sigma \in [0.1, 0.6]$. Classical: exact MLE on log-returns. Reference: conjugate posterior (Appendix G).

Cox–Ingersoll–Ross. $dX = a(b - X) dt + \sigma \sqrt{X} dW$; $dt = 0.02$, 500 steps; channels as OU. Prior $a \in [0.3, 3]$, $b \in [0.3, 1.5]$, $\sigma \in [0.1, 0.5]$. Classical: Euler pseudo-likelihood MLE (Cox et al., 1985).

Double well. $dX = (\theta_1 X - \theta_2 X^3) dt + \sigma dW$; $dt = 0.01$, 1000 steps, noise strong enough for barrier crossings; channels: fast (1×80), slow (25×36). Prior $\theta_{1,2} \in [0.5, 3]$, $\sigma \in [0.4, 1.2]$. Classical: Kramers–Moyal regression (defined above).

Stochastic Lotka–Volterra. Prey/predator pair with multiplicative noise, $dx_1 = (\alpha x_1 - \beta x_1 x_2) dt + 0.1 x_1 dW_1$, $dx_2 = (\delta x_1 x_2 - \gamma x_2) dt + 0.1 x_2 dW_2$; both components observed; $dt = 0.01$, 600 steps; channels: fast (1×50), slow (15×40). Prior $\alpha, \gamma \in [0.8, 1.5]$, $\beta, \delta \in [0.4, 1.2]$. Classical: multi-start nonlinear least squares (defined above).

FitzHugh–Nagumo. $\dot{v} = v - v^3/3 - w + I + \text{noise}$, $\dot{w} = \epsilon(v + a - bw)$; only v observed at 25 evenly spaced times over horizon 40 ($dt = 0.05$), observation noise 0.05. Prior $a, b \in [0.5, 0.9]$, $\epsilon \in [0.05, 0.3]$, $I \in [0, 0.5]$. Classical: multi-start nonlinear least squares (defined above).

SEIR with mortality. Five-compartment stochastic epidemic (two-phase contact rate $\beta_1 \rightarrow \beta_2$, incubation α , recovery γ_r , mortality γ_d); infected and deceased compartments observed at 20 days over a 60-day horizon ($dt = 0.1$), observation noise 0.004. Prior $\beta_1 \in [0.2, 0.6]$, $\beta_2 \in [0.05, 0.3]$, $\alpha \in [0.2, 0.5]$, $\gamma_r \in [0.05, 0.2]$, $\gamma_d \in [0.005, 0.05]$. Classical: nonlinear least squares on (I, D) .

Polynomial-drift SDE. $dX = (c_0 + c_1 X + c_2 X^2 + c_3 X^3) dt + \sigma dW$ — drift identification on a dense polynomial basis, the setting of SINDy (Brunton et al., 2016) without its sparsity prior (a sparse-coefficient variant is discussed in Sec. 9); $dt = 0.01$, 1000 steps; channels: fast (1×80), slow (20×45). Prior $c_0, c_2 \in [-0.5, 0.5]$, $c_1 \in [-2, 0]$, $c_3 \in [-1, -0.1]$, $\sigma \in [0.3, 0.8]$. Classical: least squares of the Kramers–Moyal drift estimate on the polynomial basis — the dense-basis version of SINDy regression.

Heston (random designs). $dS = \mu S dt + \sqrt{v} S dW_1$, $dv = \kappa(\bar{v} - v) dt + \xi \sqrt{v} dW_2$, $\text{corr}(dW_1, dW_2) = \rho$ (Heston, 1993); price observed, volatility latent; $dt = 0.01$, 500 steps, designs of $K \in [2, 128]$ random times. Prior $\mu \in [-0.1, 0.3]$, $\kappa \in [0.5, 5]$, $\bar{v} \in [0.02, 0.3]$, $\xi \in [0.1, 1]$, $\rho \in [-0.9, 0]$.

Merton jump-diffusion (random designs). $dS/S = \mu dt + \sigma dW + d(\sum_i (e^{J_i} - 1))$ with Poisson jump rate λ and $J_i \sim \mathcal{N}(\mu_J, s_J^2)$ (Merton, 1976); $dt = 0.01$, 500 steps, $K \in [2, 128]$. Prior $\mu \in [-0.1, 0.3]$, $\sigma \in [0.1, 0.5]$, $\lambda \in [0.2, 3]$, $\mu_J \in [-0.3, 0.3]$, $s_J \in [0.05, 0.4]$. Reference: Poisson-mixture transition density (Appendix G).

Hodgkin–Huxley (random designs). Standard four-dimensional membrane model (Hodgkin and Huxley, 1952); voltage observed at random times; parameters $(g_{\text{Na}}, g_K, g_L, I)$ with prior $[60, 180] \times [10, 50] \times [0.05, 0.5] \times [4, 12]$; 3000 integration steps, $K \in [4, 96]$.

Pharmacokinetics (random designs). The standard one-compartment oral-dosing model of pharmacokinetics (e.g. Gabrielsson and Weiner, 2000): absorption at rate k_a , elimination at rate k_e , distribution volume V , giving the Bateman concentration curve, concentration $c(t) = \frac{D_0 k_a}{V(k_a - k_e)} (e^{-k_e t} - e^{-k_a t})$ with dose $D_0 = 500$; log-normal assay noise (10%); horizon 24 h, $K \in [3, 64]$ irregular draws. Prior $k_a \in [0.5, 4]$, $k_e \in [0.05, 0.4]$, $V \in [20, 100]$. Reference: log-normal likelihood MCMC.

Fisher–KPP (random designs). $\partial_t \theta = D \partial_{xx} \theta + r \theta(1 - \theta)$ on $[0, 1]$ with reflecting boundaries, 64-point finite-difference grid, Gaussian initial bump; three fixed sensors at $x = 16/63, 32/63, 48/63$ observed at random (time, sensor) pairs with noise 0.02; $dt = 0.01$, 200 steps, $K \in [4, 96]$. Prior $D \in [0.001, 0.01]$, $r \in [2, 10]$. Reference: Gaussian observation model around the deterministic solve; one likelihood evaluation is one PDE solve.

Controls. Two model-free controls accompany every result. The *prior-mean predictor* ignores the data entirely (exactly 25% mean absolute error under a uniform prior). The *ridge control* is a point predictor of the parameters: a degree-2 ridge regression on generic summary statistics of the token set — per token feature, the mean, standard deviation, mean square, mean absolute value, and the 0.1/0.5/0.9 quantiles, computed over however many observations are present. It recovers no distribution and is not meant to; its role is to detect a specific silent failure that the calibration screen cannot see: a posterior that ignores the data and returns the prior passes SBC perfectly, and only a point-accuracy control unmasks it.

Calibration of the fixed-grid gallery. Per-parameter SBC p -values for the standard-budget gallery run (40,000 simulations, the run of Figs. 7 and 8) are listed in Table 5; Fig. 8 shows empirical-vs-nominal coverage of the central credible intervals, and Fig. 8 the underlying rank distributions.

system	per-parameter SBC p -values					
linear-Gaussian	m_1 : 0.06	m_2 : 0.60	m_3 : 0.38	m_4 : 0.56		
Ornstein-Uhlenbeck	θ : 0.28	σ : 0.77				
SEIR	β_1 : 0.33	β_2 : 0.22	α : 0.61	γ_r : 0.34	γ_d : 0.53	
geometric Brownian motion	μ : 0.26	σ : 0.44				
Cox-Ingersoll-Ross	a : 0.65	b : 0.001	σ : 0.73			
double well	θ_1 : 0.78	θ_2 : 0.54	σ : 0.88			
stochastic Lotka-Volterra	α : 0.07	β : 0.69	δ : 0.08	γ : 0.35		
FitzHugh-Nagumo	a : 0.19	b : 0.15	ϵ : 0.90	I : 0.63		
polynomial-drift SDE	c_0 : 0.06	c_1 : 0.96	c_2 : 0.46	c_3 : 0.80	σ : 0.50	

Table 5: Per-parameter SBC uniformity p -values for the fixed-design gallery run of the standard-budget gallery run (500 observation sets $\times$ 200 draws; $p > 0.05$ passes). The single failure is marked in bold; see Sec. 7.2 for its behavior across retrainings.

B Optimizer and stopping-rule measurements

Setup. Every number in this appendix comes from one configuration, so that only the optimizer variables change: the design-amortized GBM system, the big model (2.3M parameters), and scoring by FID_σ against the exact conjugate reference on a fixed six-bucket evaluation set (design sizes $K = 2, 5, 10, 20, 50, 95$; 32 observation sets per bucket; two-chain floors 0.0012–0.0019). Reported values are means over the six buckets. Each training run occupies one H200 GPU; several comparisons hold the example budget (batch $\times$ steps) fixed on purpose, and state it.

Step size at the reference batch. At batch 512 and 9,000 steps (4.6M examples):

Muon step η	3×10^{-4}	10^{-3}	3×10^{-3}	10^{-2}	2×10^{-2}	6×10^{-2}
FID_σ	0.0133	0.0070	0.0065	0.0061	0.0085	diverges

The basin is broad and shallow between 3×10^{-3} and 10^{-2} , and the edge beyond it is hard. The default sits at 10^{-2} .

Scaling with batch size. Repeating the sweep at batches 64, 2,048, and 8,192 (example budget fixed within each comparison) locates the optimum at 3.5×10^{-3} , 10^{-2} (flat to 2×10^{-2}), and 2×10^{-2} respectively. From batch 64 to 512 the optimum moves by $\times 2.82$ against $\times 2.83$ predicted by $\sqrt{B}$; the fourth-root alternative is excluded rather than disfavored, since its prediction costs 0.0079 against 0.0066 at batch 64. Past batch $\sim 2,048$ the optimum stops moving: at 8,192 the square root would extrapolate to 4×10^{-2} , which scores worse than the capped 2×10^{-2} (0.0060 against 0.0052), and twice the extrapolation diverges. Hence the rule quoted in Sec. 4: square root below the critical batch, constant above it.

Transfer across width. At batch 512 and fixed budget, the step 10^{-2} against the hotter 2×10^{-2} across three widths: tiny 0.0089 vs 0.0087, small 0.0066 vs 0.0076, big 0.0061 vs 0.0085. Run hot, the capacity ladder is non-monotone (big worse than small); at the law step it is ordered as capacity says it should be. “Wider is worse” can be a statement about one number inside the optimizer rather than about capacity.

Steps against examples. At a fixed budget of 9.2M examples, batch $512 \times 18,000$ steps reads 0.0051, batch $2,048 \times 4,500$ reads 0.0045, and batch $8,192 \times 1,125$ collapses to 0.0076; at the same batch and step size, doubling the steps to 2,020 recovers 0.0060. Above a step floor the example budget governs quality; below it nothing does. The floor is per-system and spans an order of magnitude: the pharmacokinetic validation curve reaches its optimum near 66,000 steps at batch 512, where GBM’s shows no interior optimum up to 36,000 — the fact that motivates validated stopping in place of per-system budgets.

Validated stopping. The monitor’s resolution must be evaluated at the monitor’s own sample count: the GBM validation set resolves 0.0013 at 4,000 draws but only 0.0044 at 1,000. Both failure modes of the naive keep-the-best rule were measured before the margin was adopted. A monitor coarser than its model ships the deepest noise dip: 0.0053 on test against 0.0040 for the final weights. A resolving monitor still ships 0.0052, because under a cosine schedule mid-run checkpoints are systematically inflated, and an

		tiny (0.1M)	small (0.5M, default)	big (2.3M)
GBM (exact conjugate)	20,000 sims	0.039	0.039	0.037
	120,000	0.016	0.021	0.024
	480,000	0.010	0.012	0.018
	external tools	raw data 2.3–8.5; hand-derived inputs 0.34–1.5		
Ornstein–Uhlenbeck (exact-lik. MCMC)	20,000	0.091	0.070	0.079
	120,000	0.042	0.055	0.054
	480,000	0.028	0.029	0.039
	external tools	1.2 (FMPE) – 2.7 (NPE)		
Fisher–KPP, $K = 40$ (PDE-lik. MCMC)	20,000	0.277	0.118	0.084
	120,000	0.053	0.036	0.045
	external tools	0.5 (NPE); 76 (FMPE)		
Merton, $d = 5$ (mixture-lik. MCMC)	120,000	0.074	0.052	0.051
	480,000	0.033	0.035	0.042
	external tools	— (random designs; fixed-length interfaces do not apply)		
pharmacokinetics, $d = 3$ (log-normal MCMC)	120,000	0.063	0.044	0.038
	external tools	—		

Table 6: The main grid (Adam recipe, Sec. 4; evaluation sets of Sec. 7.1): median FID of the same architecture at three sizes (columns) and increasing simulation budgets (rows), for every task with a validated reference; cells average over training seeds where several were run, and the big column uses the width-scaled learning rate. At $d = 2$ (GBM, OU) the three sizes agree within seed spread — Table 8 gives the per-seed ranges — so the small differences across a size row are not significant there; capacity begins to matter only at $d \geq 3$. GBM, OU, and Fisher–KPP use the fixed-design protocol shared with the external tools, which enter the same table at the 20,000-simulation budget with their package defaults (ranges over `sbi` NPE/FMPE, BayesFlow 2, and the Simformer port; Sec. 7.4 attributes the failures). Merton and pharmacokinetics are trained and evaluated in the random-design mode on their fixed evaluation sets, scored against the converged references of Sec. 5.2 (two-chain floors 0.0065 and 0.0023). Pharmacokinetics is shown at 120,000 only: its 32-instance evaluation set has a heavy right tail (mean 2–3 $\times$ the median), and the per-set median at 480,000 swings between 0.04 and 0.24 across step counts without a monotone trend, so no stable point exists to report. External tools consume fixed-length vectors and have no entry on the two random-design systems.

argmin over the whole curve mistakes that for a reversal. The margin rule — accept the best checkpoint only when it beats the final weights by more than the monitor’s resolution — passes both cases, and a genuine late reversal clears it by an order of magnitude: on pharmacokinetics the validation curve turns from 0.1066 at step 66,222 to 0.1321 at 72,072, twenty floors, and the selected checkpoint recovers the difference.

C Adam-recipe control ladders

The tables below were measured under the Adam recipe (Sec. 4); they are referenced from Sec. 7.5 and share the fixed-design protocol that the external tools of Sec. 7.4 consume.

D Full input-representation ladder on GBM

Every arm of the GBM comparison, including the hand-engineered input variants discussed in Sec. 7.4.

E Encoder ablations for design amortization

The ablations of this appendix track the posterior width component rather than FID: the mechanism under study — set-size blindness of normalized attention — expresses itself specifically as width error, and FID would mix it with unrelated deviations. Table 10 compare encoder variants for design amortization on GBM, trained at $K \sim \text{log-uniform}[2, 128]$ random-time designs and evaluated against the exact per-design posterior (96 observation sets per bucket). Three observations organize the table. First, index-based positions combined with in-place subsampling during training (leaving kept tokens at spread positions while inference packs contiguously) produce biases of +1.1 to +3.3 posterior sd with no individually

system	trajectories	FID at epochs			
		10	35	120	120 + subsampling
GBM	10,000	0.105	0.059	0.965	0.114
	30,000	0.036	0.047	0.659	—
	120,000	0.044	0.026	0.131	0.277
OU	10,000	0.206	0.151	1.892	0.253
	30,000	0.090	0.095	1.648	—
	120,000	0.038	0.066	0.648	0.143

Table 7: Median FID when the same fixed-design training is given more simulated trajectories (rows) or more passes over them (columns), at the default model size; cells on an anti-diagonal cost equal compute. The last column repeats the 120-epoch run with a fresh random 75% token subset at every optimizer step.

model (width $\times$ blocks)	params	GBM	OU	pharmacokin.	Merton
		$d = 2$	$d = 2$	$d = 3$	$d = 5$
pico (8×2)	7,114	0.028–0.029	0.052–0.095	0.675	0.243
nano (16×2)	25,730	0.021–0.027	0.042–0.054	0.167	0.091
tiny (32×2)	97,522	0.014–0.018	0.042–0.043	0.063	0.074
small (64×3 , default)	483,346	0.018–0.026	0.045–0.066	0.044	0.052
big (128×4)	2,321,554	0.024	0.054	0.038	0.051

Table 8: Capacity ladder under the Adam recipe (all four columns; Sec. 4): median FID at a fixed budget of 120,000 simulations; sizes are given as width $\times$ transformer blocks, with parameter counts for the GBM instance (counts vary by a few percent across systems with the input dimension). Ranges span independent training seeds. GBM and OU use the fixed-design protocol of Table 7; pharmacokinetics and Merton use the design-amortized protocol, evaluated against the converged references of Sec. 5.2 whose own two-chain floors are 0.0023 and 0.0065 — every entry in those two columns is above its floor by a factor of 8 or more. The big row uses the width-rescaled learning rate (see text). The ladder is flat within seed spread at $d = 2$, and develops a capacity floor that rises with the parameter dimension — near 10^5 parameters at $d = 3$, near 5×10^5 at $d = 5$, where accuracy improves down to the default size and then saturates (big differs from small by less than the reference floor).

incorrect component — the positional pathology of Sec. 3.2; repacking or replacing positions with time phases removes the bias entirely. Second, all point-token variants share inflated widths at dense designs, in proportion to how poorly their pooled representation encodes $\log K$ (rightmost column; the probe of Sec. 3.3) — including two attention mechanisms specifically constructed to exploit continuous time (a learned per-head time-kernel logit bias in the spirit of relative-position methods, and edge-message attention carrying gap-modulated increments on all pairs). Third, pair tokens with time-phase attention, plus the training procedure of Sec. 4, bring every bucket to within 7% of the exact width.

F Choice of discrepancy measure

Equation (10) is the squared 2-Wasserstein distance between Gaussian fits, and the division by the reference standard deviation is ours, not part of that distance: the summands carry different physical units — a rate in h^{-1} , a volume in litres — so some normalization is required, and the one we chose contracts with the posterior itself. Table 11 therefore evaluates five discrepancies on the same sample sets: the Gaussian W_2^2 under two different normalizations, two distribution-free alternatives, and the classifier two-sample accuracy used as the primary metric of the **sbibm** benchmark (Lueckmann et al., 2021). Three readings. The distribution-free measures agree with the Gaussian one on every system, so the Gaussian summary is not what limits the comparison — the normalization is: the same model reads 0.009 on GBM and 0.58 on dense pharmacokinetics in posterior-sd units, but 0.0001 against 0.0003 in prior-range units. C2ST is the most readable of the five, being bounded and calibrated at 0.5: five of the six systems sit at 0.52–0.55, i.e. a trained classifier can barely separate our posterior samples from exact ones, with dense pharmacokinetics the single exception at 0.77. And the null column earns its place — it is what exposed the under-converged Merton reference of Sec. 5.2.

method	input	train	inference	FID
amortix CFM	raw prices	236 s	70 ms	0.04
sbi NPE	raw prices	170 s	36 ms	4.8
sbi FMPE	raw prices	215 s	359 ms	3.8
BayesFlow 2 coupling	raw prices	1252 s	0.3 ms	2.6
BayesFlow 2 flow matching	raw prices	790 s	95 ms	2.3
Simformer port	raw prices	703 s	1049 ms	8.5
sbi NPE	log-prices	157 s	25 ms	5.8
sbi NPE	log-returns	52 s	26 ms	1.5
sbi FMPE	log-returns	99 s	378 ms	0.8
BayesFlow 2 coupling	log-returns	1254 s	0.3 ms	1.2
BayesFlow 2 flow matching	log-returns	771 s	97 ms	1.4
Simformer port	log-returns	586 s	1053 ms	0.34
exact-likelihood MCMC	log-returns (analytic)	—	42 ms (CPU)	reference

Table 9: GBM: all methods at package defaults, one H200 GPU, equal 20,000-simulation budget, 200 test observation sets against the exact conjugate posterior. FID is the Fréchet distance (10) to the reference, median over test observation sets (the distribution across sets is right-skewed; means are 1.5–3× larger, per-arm values in **results/**). Middle block: raw data, the amortized default regime. Bottom block: the same external tools given hand-derived input coordinates. Inference time is per observation set for 2,000 samples.

encoder variant	σ width ratio at K				$R^2(\log K)$
	5	9	20	100	
point tokens, index positions, spread subsampling	0.94*	1.13*	1.42*	2.08*	—
point tokens, index positions, contiguous	1.05	1.17	1.13	1.73	0.957
point tokens, time-kernel attention bias	1.03	1.14	1.31	2.51	0.712
point tokens, edge-message attention	1.03	1.15	1.29	2.45	0.498
point tokens, time phases	1.01	1.10	1.14	1.95	—
+ explicit $\log K$ feature	1.01	1.05	1.08	1.49	—
pair tokens, time phases	1.02	1.02	1.02	1.27	0.985
pair tokens + full training procedure	1.06	1.06	1.05	1.07	—

Table 10: GBM with variable designs: posterior width for the volatility σ against the exact per-design posterior (96 observation sets per design-size bucket) and, where measured, the probe R^2 for $\log K$ from the pooled encoder output. The table tracks σ because it is the parameter whose posterior width is governed by the observation density — the drift μ is weakly informed at every K , and all variants recover its posterior within a few percent, so it does not discriminate between architectures. *This row also carries biases of +1.09 to +3.30 posterior sd (the positional train/test mismatch of Sec. 3.2); all other rows have biases consistent with zero.

G Reference posteriors

GBM conjugate posterior. With observations at indices $i_0 = 0 < i_1 < \dots$ of the fine grid, log-returns $r_k = \log(S_{i_{k+1}}/S_{i_k})$ over gaps τ_k are independent $\mathcal{N}((\mu - \sigma^2/2)\tau_k, \sigma^2\tau_k)$. In the coordinates $(b, \sigma^2) = (\mu - \sigma^2/2, \sigma^2)$ the model is a Gaussian location–scale family with sufficient statistics $T = \sum_k \tau_k$, $R_1 = \sum_k r_k$, and $SS_w = \sum_k r_k^2/\tau_k - R_1^2/T$; draws from the standard conjugate posterior ($\sigma^2 = SS_w/\chi_{n-1}^2$, $b \mid \sigma^2 \sim \mathcal{N}(R_1/T, \sigma^2/T)$) are transformed to (μ, σ) and importance-resampled to the uniform box prior with the appropriate Jacobian weight. The sampler was validated against exact-likelihood MCMC on the same designs: over 24 comparisons, worst posterior-mean discrepancy 0.099 sd, width ratios 0.91–1.03.

Adaptive Metropolis. Random-walk Metropolis on the box, proposal $x' = x + sC\xi$, $\xi \sim \mathcal{N}(0, I)$. During burn-in (default $\max(500, n/2)$ iterations) the scalar s is adapted toward a 0.25 acceptance rate in blocks of 50 with a decaying Robbins–Monro gain, and once sufficient burn-in states are collected the shape C is replaced by the Cholesky factor of the empirical covariance rescaled by $2.38/\sqrt{d}$ (Haario et al., 2001); adaptation is frozen for the retained phase. Out-of-box proposals are rejected, which samples the box-truncated target exactly since the proposal is symmetric. Likelihoods: per-gap Euler–Maruyama Gaussian transitions for OU (matching the generative process, plus the stationary density of X_0); the exact GBM log-return density; the Poisson-mixture transition density for Merton (truncated

system	W_2^2 Gaussian		distribution-free		
	/ post. sd	/ prior range	energy	sliced W_2^2	C2ST
geometric Brownian motion	0.009	0.0001	0.004	0.005	0.520
linear-Gaussian	0.043	0.0001	0.011	0.011	0.548
Ornstein–Uhlenbeck	0.050	0.0008	0.027	0.028	0.551
Fisher–KPP	0.071	0.0000	0.023	0.031	0.546
Merton jump-diffusion	0.035	0.0008	0.008	0.007	0.522
pharmacokinetics	0.580	0.0003	0.191	0.182	0.769
<i>null: the same statistic between two independent reference draws</i>					
geometric Brownian motion	0.002	0.0001	0.001	0.002	0.500
linear-Gaussian	0.004	0.0000	−0.001	0.002	0.498
Ornstein–Uhlenbeck	0.013	0.0003	0.003	0.009	0.507
Fisher–KPP	0.006	0.0000	0.004	0.008	0.510
Merton jump-diffusion	0.007	0.0003	0.000	0.002	0.497
pharmacokinetics	0.028	0.0000	0.008	0.014	0.527

Table 11: Five discrepancies between the same trained model and the same references, on all six reference systems, medians over 16–32 test observation sets with 4,000 samples each, and (lower block) the floor each statistic carries at that sample size. Energy and sliced W_2^2 are computed in posterior-sd units like the first column. The measures rank the systems identically; what separates them is the normalization, not the Gaussian assumption. Merton is scored against the converged reference of Sec. 5.2 (its 4,000-draw chains fail the two-chain check, which is why the floor column matters).

at the numerically negligible jump count); log-normal residuals around the Bateman curve; a Gaussian observation model around the deterministic PDE solve for Fisher–KPP. Every chain used as a reference is paired with a second chain from a well-separated start; between-chain posterior-mean discrepancies are reported alongside the results they certify (Sec. 7.4).

H Correspondence with package options

in this report	in the package
point tokens	<code>embed="wpoint"</code>
pair tokens + set conditioning	<code>embed="wfilm"</code>
warped values/increments (fixed designs)	<code>embed="wbasis" / "wdiff"</code>
attention phases from physical time	<code>rope="time"</code>
fresh designs each optimizer step	<code>fit(retokenize=...) via make_retokenizer()</code>
Muon with the batch step law	<code>fit(optimizer="muon"); muon_lr=None</code> → the measured law
validated stopping with margin	<code>fit(keep_best=True, monitor_floor=...)</code>
evaluation-set resolution at n draws	<code>evaluation.floor_at</code>
variable-design problem wrapper	<code>amortix.designs.DesignProblem</code>
benchmark systems	<code>amortix.problems (DESIGN_Z00)</code>
calibration screen / reference machinery	<code>sbc_design / amortix.mcmc + likelihood factories</code>
comparison scripts (Sec. 7.4)	<code>examples/baseline_{npe,ou,kpp,bayesflow,simformer}.py</code>

All defaults named in this report are selected automatically by `FlowPosterior(problem)`; the non-default options exist as reproducible controls for the comparisons above.

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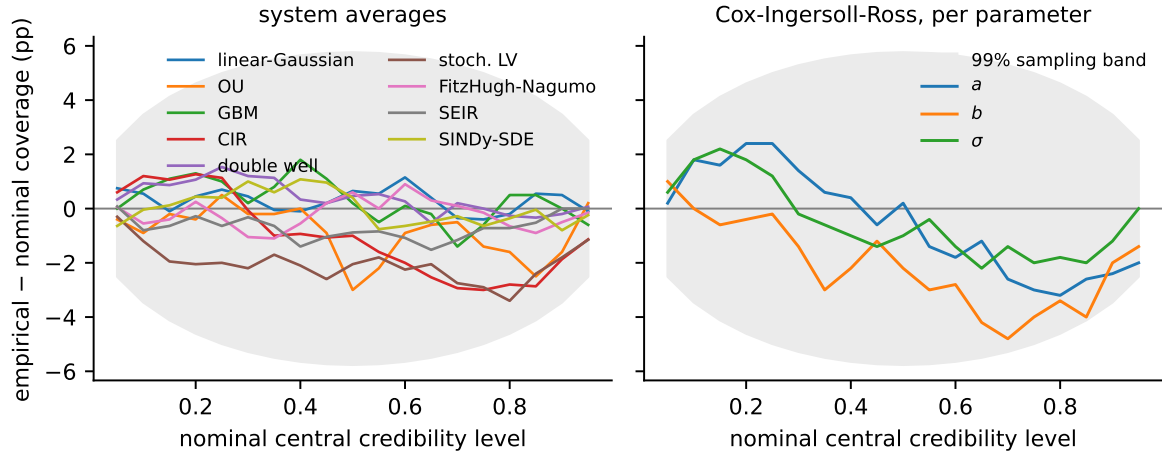


Figure 7: Empirical coverage of central credible intervals for the fixed-design gallery, shown as the deviation from the nominal level. For each system, 500 synthetic observation sets are simulated with known parameters and the posterior’s central $q\%$ interval is checked against the truth; a calibrated posterior has zero deviation at every level, and the gray band is the 99% pointwise range induced by the finite number of test observation sets. Left: deviation averaged over each system’s parameters. Right: individual parameters of Cox–Ingersoll–Ross, the system with the largest deviation in this run; the b curve reflects the ≈ 0.15 sd location shift discussed in the text. Rank-level detail for all 32 parameters is in Appendix A (Fig. 8).

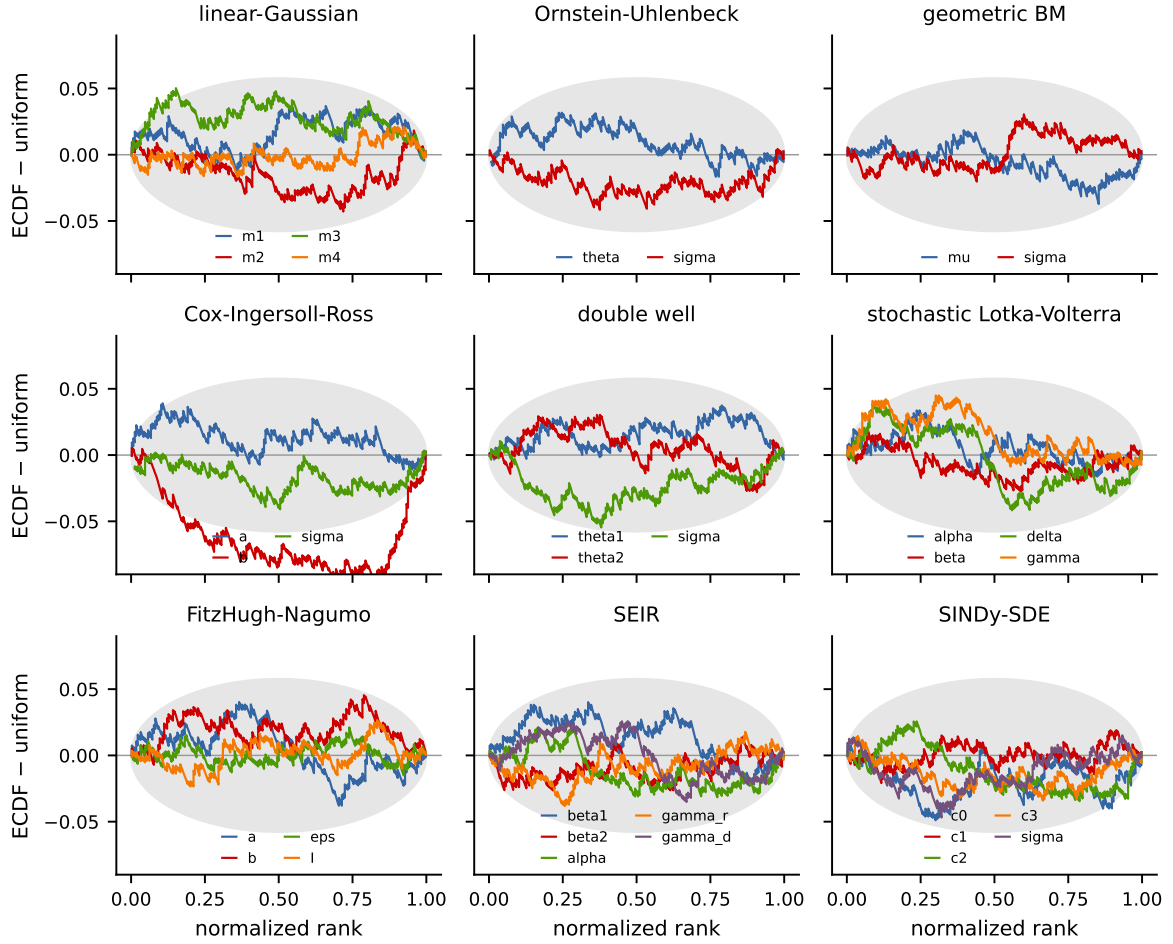


Figure 8: Calibration check for the fixed-design gallery. Procedure: for each system, 500 synthetic observation sets are simulated with known parameters; for each observation set the model produces 200 posterior draws, and the rank of the true parameter among the draws is recorded. If the posterior is calibrated, these ranks are uniform. Each curve shows, for one parameter, the deviation of the empirical CDF of normalized ranks from the uniform diagonal; the gray band is the 99% range attributable to sampling noise at 500 observation sets. A curve bulging below zero on the left indicates a posterior shifted low relative to the truth, and vice versa. All curves stay within the band except Cox–Ingersoll–Ross b , whose deviation corresponds to the ≈ 0.15 sd location shift discussed in the text.